Network Meta-analysis

Georgia Salanti, Deborah Caldwell, Anna Chaimani, and Julian Higgins

Abstract

The increasing number of alternative treatment options for the same condition created the need to undertake reviews that address complex policy-relevant questions and make inferences about many competing treatments. Such reviews collect data which, under conditions, can be statistically synthesized using network meta-analysis. This chapter presents the basic concepts of indirect and mixed comparison of treatments and presents the statistical models for network meta-analysis and their implementation both theoretically and in examples. The assumption underlying network meta-analysis is extensively discussed and extensions of the models to account for effect modifiers are presented.

G. Salanti (✉) • A. Chaimani
Department of Hygiene and Epidemiology, University of Ioannina School of Medicine, Ioannina, Greece
e-mail: georgia.salanti@ispm.unibe.ch; annachaimani@gmail.com

D. Caldwell
School of Social and Community Medicine, University of Bristol, Bristol, UK
e-mail: d.m.caldwell@bristol.ac.uk

J. Higgins
MRC Biostatistics Unit, Cambridge, UK

Centre for Reviews and Dissemination, University of York, York, UK
e-mail: julian.higgins@bristol.ac.uk

Contents

Introduction 1
Example: Incident Diabetes with Antihypertensive Drugs 2
A Roadmap to the Chapter 2

Meta-analysis of Head-to-Head Comparisons 3
Types of Data that Feed into a Meta-analysis 3
Meta-analysis and Meta-regression as Linear Model 4
Meta-analysis as Hierarchical Model 5
Fitting the Meta-analysis Model 5

Indirect and Mixed Comparison 7
Theory and Formulae for Indirect Comparisons 7
Theory and Formulae for Mixed Comparisons 9
Assumptions Underlying Indirect and Mixed Comparisons 10

Models for Network Meta-analysis 15
Consistency Models 15
Assumptions of Network Meta-analysis 26
Statistical Methods to Detect Inconsistency in a Network of Interventions 26
Inconsistency Models 29
Exploring Heterogeneity and Inconsistency: Network Meta-regression 31

Numerical and Graphical Presentation of Results from Network Meta-analysis 35

References 37

Introduction

Meta-analyses of randomized controlled trials (RCTs) are often considered to provide the most reliable and valid evidence on which to base healthcare decisions, usually ranking above single

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RCTs in evidence-based medicine (EBM) hierarchies of evidence (Higgins and Green 2008). Meta-analysis is an integral part of EBM, used by international health organizations such as the World Health Organization and The Cochrane Collaboration, and is widely used to inform health technology assessment and clinical guidelines produced by organizations such as the Canadian Agency for Drugs and Technologies in Health (CADTH), the National Institute for Health and Clinical Excellence (NICE), and the Agency for Healthcare Research and Quality (AHRQ). However, as meta-analysis traditionally compares two treatments at a time (a pairwise comparison) its use in the presence of multiple competing treatment options is limited. Moreover, although clinical and policy making interest lies in comparing active agents, new drugs are commonly compared with placebo in order to obtain marketing authorization. Given that clinical practice changes over time and that licensed or reference treatments differ across countries, it is unrealistic to expect that individual trials and pairwise meta-analyses can provide evidence of comparative effectiveness for every intervention of interest.

The need to compare multiple competing treatments to inform clinical guidelines and health technology appraisals has underpinned the development of network meta-analysis. Also known as a multiple treatment meta-analysis and mixed treatment comparisons, a network meta-analysis simultaneously combines direct and indirect information across a network of studies to make inferences regarding the relative effectiveness of multiple interventions. An indirect comparison, which underpins the method, is a simple idea: treatment. A can be compared with treatment B via a common comparator C, by statistically combining the comparison A versus C (AC) and B versus C (BC) studies. Several applications and methods papers have outlined the benefits of combining direct and indirect evidence in a network meta-analysis (Caldwell et al. 2005; Cooper et al. 2011; Hoaglin et al. 2011; Mills et al. 2011). These include improvement in precision for the estimated effect sizes and the ability to compare treatments that have not been directly compared in any trial. Despite the increasing number of applications, network meta-analysis is far from being an established practice. Many authors emphasize the secondary or supplementary nature of the analyses, giving priority to direct evidence (NICE 2008; Edwards et al. 2009). Network meta-analyses are often considered controversial (Piccini and Kong 2011; Thijs et al. 2008), for example, a recent evaluation of the relative effectiveness of twelve new-generation antidepressants attracted supporters as well as skeptics (Barbui et al. 2009; Cipriani et al. 2009).

Example: Incident Diabetes with Antihypertensive Drugs

To exemplify all methodologies of this chapter, a published network meta-analysis (Elliott and Meyer 2007) will be used. It is based on a systematic review that aimed to compare antihypertensive drugs with respect to the incidence of diabetes. The review included 22 randomized controlled trials. The competing interventions are placebo (P), β-blockers (BB), diuretics (D), calcium channel blockers (CCB), angiotensin converting enzyme (ACE) inhibitors and angiotensin II receptor blockers (ARB).

Four studies include three arms and the rest are two-arm trials. All comparisons have been evaluated in at least one study except for the comparisons ARB versus ACE inhibitors for which no studies exist. Figure 1 shows the plot of this network of intervention comparisons.

A Roadmap to the Chapter

The chapter starts by setting up notation and the two most commonly applied models used for meta-analysis and meta-regression of pairwise comparison and discusses the frequentist and Bayesian implementation of the models (section "Meta-analysis of Head-to-Head Comparisons"). Section "Indirect and Mixed Comparison" describes the theory of indirect comparisons and the combination of these with direct comparisons (sometimes called "mixed" comparisons) in a simple three-treatment network consisting of

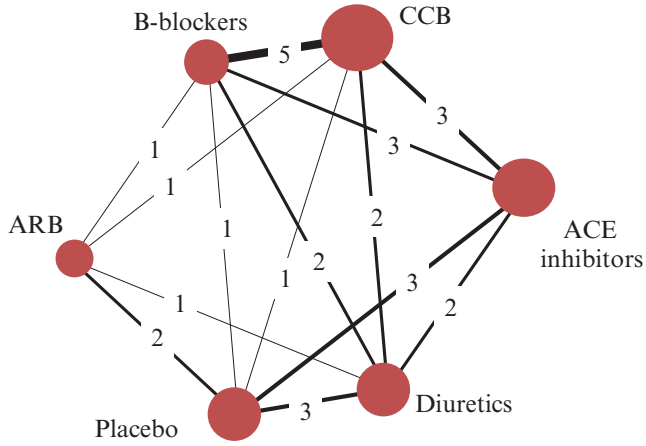


Fig. 1 Plot of network for incidence of diabetes. The size of the nodes is proportional to the number of studies that evaluate each intervention and the thickness of the lines is proportional to the frequency of each comparison in the network. The numbers represent the number of studies including each comparison (*CCB* calcium channel blockers, *ARB* angiotensin receptor blockers, *ACE* angiotensin converting enzyme)

trials that compare any two of the three treatments. The key assumption required to derive valid indirect and mixed estimates is extensively discussed in section "Assumptions Underlying Indirect and Mixed Comparisons." Section "Models for Network Meta-analysis" is more technical and describes the models used to fit network meta-analysis and discusses statistical methods to detect and account for violation of the key assumption. Section "Models for Network Meta-analysis" concludes by outlining extensions of network meta-analysis to account for the impact of effect modifiers. Section "Numerical and Graphical Presentation of Results from Network Meta-analysis" reviews numerical and graphical methods for presenting results from network meta-analysis to assist clinicians with interpretation of findings.

Meta-analysis of Head-to-Head Comparisons

Pairwise meta-analysis summarizes the relative effectiveness of two interventions across *N* studies. Two basic parametric models are usually used: the fixed-effect model and the random effects model. Under the fixed-effect assumption, it is considered that all studies estimate the same underlying treatment effect. In the random effects model, it is assumed that there is a study-specific treatment effect underlying each study and that the observations from different studies estimate these different underlying effects. The study-specific underlying effects can be different yet related, and it is assumed that they "belong" to the same distribution. The variance of this distribution is the heterogeneity parameter describing the magnitude of the between-study variation. Meta-analysis can be viewed as special case of a weighted linear regression or as a hierarchical model. Both models are equivalent; though linear regression approaches are the most common approach in a frequentist implementation when treatment effect estimates are the starting point of the analysis (known as a "contrast-based" approach), and hierarchical approaches are usually encountered when summary data from each treatment group are the starting point of the analysis (an "arm-based" approach), often fitted in a Bayesian framework. These ideas are discussed in detail in the following three subsections.

Types of Data that Feed into a Meta-analysis

The systematic review process first requires identification and appraisal of studies that address the

research question of interest. Then, relevant data are extracted from the studies that fulfill the predefined inclusion criteria. Consider that N studies, indexed with $i = 1, \ldots, N$ and comparing two treatments. A and B are included that contribute data for a particular outcome of interest. The data from each study can be *arm based* or *contrast based.* The first term refers to data that apply to each arm; for dichotomous outcomes these can be the number of successes r_{iA}, r_{iB} out of the total randomized n_{iA}, n_{iB} for the arms A and B, respectively. For continuous outcomes, the arm-based data are the outcome means m_{iA}, m_{iB}, standard deviations sd_{iA}, sd_{iB} and total numbers of participants n_{iA}, n_{iB} per arm.

Instead of presenting the outcome in each arm separately, a study can report the difference in the outcome between the two arms using a statistic. The *contrast-based* approach refers to study-specific statistics that compare the two arms. With dichotomous outcomes, the statistics are usually the odds ratios (OR), risk ratios, risk differences, or hazard ratios, whereas for continuous outcomes, it is usually mean differences, standardized mean difference, or ratios of means. The logarithmic transformation of ratio measures (e.g., odds and risk ratios) is typically applied in practice. Let y_{iAB} be generic notation for one of these statistics, which will be referred to as "the effect size." The sample variance of an effect size will be denoted with s^2_{iAB} Of course, arm-specific data can be transformed into contrast-based data before the start of the analysis. However, modeling arm-specific data is often an advantage in terms of model fit and therefore detailed data, if available, should be given preference.

Meta-analysis and Meta-regression as Linear Model

Meta-analysis can be viewed as a linear regression model with no covariates. As each observation represents a study and these studies typically have different sample sizes, it is reasonable to weight the observations accordingly; hence, meta-analyses are fitted as a weighted linear model. In a random effects meta-analysis, a study's effect size is given by

$$y_i = \mu + \delta_i + e_i \tag{1}$$

where μ is the summary treatment effect and the random errors are assumed normally distributed

$$e_i \sim N\left(0, s_i^2\right) \tag{2}$$

The quantities δ_i account for random variation in the treatment effects across studies (heterogeneity) and are assumed to be normally distributed as $\delta_i \sim N(0, \tau^2)$. Setting either the heterogeneity variance τ^2 to be zero or all $\delta_i = 0$ reduces the model to the fixed-effect model.

Equation 1 can be extended into meta-regression in order to account for variability in the effect sizes with respect to a trial-specific variable x_i:

$$y_i = \mu_1 + \mu_2 x_i + \delta_i + e_i$$

When x_i is a categorical variable then the meta-regression model is equivalent to subgroup analysis. Consider, for example, that the systematic review comprises studies with appropriate and inappropriate blinding (subgroups 1 and 2). Then, using the dichotomous variable x_i as an index variable which takes values 0 for appropriate and 1 for inappropriate blinding, a subgroup analysis via meta-regression can be fitted. Then, the summary effect μ_1 would be the summary estimate from the subgroup of appropriately blinded studies and $\mu_1 + \mu_2$ would be the summary estimate from the subgroup of inappropriately blinded studies. In a general framework of F subgroups indexed with f, a meta-regression model can be fitted without intercept:

$$y_i = \sum_{f=1}^{F} \mu_f x_{if} + \delta_i + e_i \tag{3}$$

where now the regression coefficients μ_f are the summary estimates in subgroups.

Meta-analysis as Hierarchical Model

An alternative representation of the random effects meta-analysis model is to consider two levels of estimation hierarchy: one level for the observation in each study that estimates the study-specific underlying effect and a second level for all the study-specific underlying effects that arise form a common distribution centered around the meta-analysis summary effect. Specifically, in each study the observed effect size y_i is assumed normally distributed with mean equal to the underlying effect size θ_i, and uncertainty reflected by the sample variance:

$$y_i \sim N\left(\theta_i, s_i^2\right) \tag{4}$$

Then, it is assumed that the underlying θ_i form a common distribution with expectation The μvariance of the distribution is the heterogeneity:

$$\theta_i \sim N\left(\mu, \tau^2\right) \tag{5}$$

The equivalence between the two alternative representations (linear and hierarchical model) is seen by identifying θ_i with $\delta_i + \mu$. The fixed effects model can be obtained by substituting $\theta_i = \mu$ into distribution (5).

The hierarchical model presented in this section can be used to model arm-specific data instead of effect sizes. This offers the advantage that the true likelihood of the data can be used and bypasses the assumption of normality for the observed effect sizes (as reflected in distributions (2) and (4)), often yielding better fit of the models. For example, when the outcome is dichotomous the normal likelihood in Eq. 5 can be substituted by two binomial likelihoods:

$$r_{iA} \sim B(p_{iA}, n_{iA})$$

$$r_{iB} \sim B(p_{iB}, n_{iB})$$

Then the probabilities of success in the two arms can be parameterized to derive contrast-specific parameters θ_i using a link function φ:

$$\varphi(p_{iA}) = u_i$$

$$\varphi(p_{iB}) = u_i + \theta_i$$

For instance, the underlying study-specific treatment effect θ_i can be the log-odds ratio if φ is the logit function or the log-risk ratio if φ is the logarithmic function. For more details, see (Warn et al. 2002).

When the outcome is continuous, distribution (4) is substituted by the two normal distributions for the means in the two arms:

$$m_{iA} \sim N\left(\lambda_{iA}, (sd_{iA})^2/\sqrt{n_{iA}}\right)$$

$$m_{iB} \sim N\left(\lambda_{iB}, (sd_{iB})^2/\sqrt{n_{iB}}\right)$$

Then, the effect size can be derived by parameterizing the two means λ_{iA} and λ_{iB}; for example, the mean difference could be derived as $\theta_i = \lambda_{iA} - \lambda_{iB}$

For either type of data (dichotomous or continuous) and for any statistic, the underlying effects θ_i are assumed to arise from a common distribution as in (5).

Fitting the Meta-analysis Model

The meta-analysis models above can each be fitted within a frequentist or a Bayesian framework. This section briefly summarizes the practical differences between the two approaches and the implications they might have for the summary estimates. For a more detailed overview of the Bayesian methodology, the reader should refer to Spiegelhalter et al. (2004) and Sutton and Abrams (2001). The choice between the different frameworks depends primarily on familiarity with the required software and methods.

The main practical differences between frequentist and Bayesian implementations relate to how the methods estimate the heterogeneity. In

most frequentist implementations, the parameter τ is assumed "known" and several estimation approaches have been proposed including the method of moments and restricted maximum likelihood (see Viechtbauer 2007). Accounting for uncertainty in the estimation of heterogeneity is possible, but most existing software does not include uncertainty for τ in standard meta-analysis routines. The frequentist estimates invariably perform poorly when few studies are included in the meta-analysis. In a Bayesian framework, τ may easily be treated as a random variable and is given a prior distribution which, combined with likelihood statement, provides inference on the (posterior) distribution of the heterogeneity parameter. Therefore, uncertainty about the estimation of τ is always introduced and impacts on the results. However, with few studies, Bayesian estimation of heterogeneity is also problematic because the choice of the prior distribution may have considerable impact on the results since little information is provided from the data (Lambert et al. 2005). In such cases, it is particularly advisable to carry out sensitivity analyses.

In a Bayesian framework the fit of the model to the data can be measured by calculating the posterior mean residual deviance $\overline{D}$ The model fits the data adequately when $\overline{D}$ approximates the number of unconstrained data points (e.g., the number of studies when the contrast-based approach is used in a head-to-head meta-analysis). The deviance information criterion (*DIC*) is the sum of $\overline{D}$ and the effective number of parameters, *pD*, and provides a measure of model fit penalized for model complexity (Spiegelhalter et al. 2002). It has an interpretation similar to the Akaike information criterion: lower values of the *DIC* suggest a better compromise between model fit and complexity. A difference in *DIC* of three units or more is usually considered important. *DIC* can be used to compare different models as long as they are applied to the same amount of data. For example, *DIC* can be used to select between different meta-regression models to choose between consistency and different inconsistency models, as will be discussed later.

An advantage of the Bayesian fitting of the models is that the posterior distribution can be directly interpreted as the probability distribution of the quantity of interest (e.g., summary effect, heterogeneity). Consequently, probabilistic statements follow naturally; it is straightforward to calculate probabilities of one treatment being better than the other, or outperforming another by a specific magnitude. This is an important advantage when many treatments need to be compared and pairwise presentation of effect sizes becomes cumbersome. Calculation of probabilities is possible in a frequentist setting via resampling techniques, but this typically requires specialized routines or extra programming for the user.

Several software options exist that fit meta-analysis models in a frequentist setting. Freely available software includes RevMan and packages in R; a popular commercial option is STATA. The available routines and software frame the flexibility of models; for instance, it is not possible to fit arm-specific data using their exact likelihood with the existing meta-analysis specific routines.

With network meta-analysis increasing in popularity, Bayesian approaches have become popular as they offer greater flexibility, and WinBUGS is the most common software used. Meta-analysis can be fitted as a linear or hierarchical model and both arm-specific or contrast specific data can be modelled, giving Bayesian fitting a practical advantage compared to the frequentist approach.

Example: Subgroup Meta-analysis for ACE Inhibitors and CCB Versus β-Blockers

To exemplify the methods outlined above, consider the two comparisons CCB versus β-blockers and ACE inhibitors versus β-blockers from the network introduced earlier relating to incident diabetes. Firstly, a meta-regression model will be fitted, with dummy variables, to carry out subgroup analysis on contrast-based data (the ln(*OR*) for diabetes from each study), using the treatments being compared to define two subgroups. There are three studies comparing ACE inhibitors versus β-blockers and five comparing CCB versus β-blockers. Although a regression model is usually written with an intercept and one or more regression terms, it can also be written with no intercept as in Eq. 3. The eight observed ln(*OR*) estimatesare

denoted as y_i using study indices, $i = 1, 2, \ldots, 8$. Each y_i is then written as a function of the variables $x_{iACE-BB}$ and $x_{iCCB-BB}$. These variables take values $x_{iACE-BB} = 1$ if study i compares ACE inhibitors versus β-blockers and $x_{iACE-BB} = 0$ otherwise, and $x_{iCCB-BB} = 1$ for CCB versus β-blockers and zero otherwise. The meta-regression model that gives the summary effects for these two comparisons is

$$y_i = \mu_{ACE-BB} x_{iACE-BB} + \mu_{CCB-BB} x_{iCCB-BB} + \delta_i + e_i$$

where δ_i is the study-specific random effect. Fitting this model in STATA using the command **metareg** and specifying the method of moments as the method to estimate the heterogeneity parameter produces the results shown in the upper part of Table 1.

The coefficients μ of the regression are the subgroup-specific summary effects μ_{ACE-BB}, μ_{CCB-BB} on the ln(OR) scale. The heterogeneity parameter was estimated as $\tau^2 = 0.01$ and the proportion of variability due to heterogeneity rather than sampling error (after accounting for subgroup differences) as $I^2 = 59\%$.

The subgroup meta-analysis can also be fitted with the 2 × 2 tables as the starting point rather than the ln(OR), and it is convenient to write this implementation as a hierarchical model. The outcome in each study is the number of patients diagnosed with diabetes and therefore the binomial likelihood can be used in a hierarchical model. This means that the number of events (patients with diabetes) in each study arm, r_{iBB} and r_{iACE} for the first three studies comparing ACE inhibitors to β-blockers or r_{iBB} and r_{iCCB} for the five studies comparing CCB to β-blockers, follow a specific binomial distribution with a respective probability of success:

$$r_{iBB} \sim B(p_{iBB}, n_{iBB}), \quad i = 1, \ldots, 8.$$

$$r_{iACE} \sim B(p_{iACE}, n_{iACE}), \quad i = 1, \ldots, 3$$

$$r_{iCCB} \sim B(p_{iCCB}, n_{iCCB}), \quad i = 4, \ldots, 8$$

The log-odds ratios can be written as functions of the arm-specific probabilities; the two ln(OR) are logit(p_{iACE}) − logit(p_{iBB}) and logit(p_{iCCB}) − logit(p_{iACE}). In this case the parameterization of the model is:

$$\text{logit}(p_{iBB}) = u_i$$

$$\text{logit}(p_{iACE}) = u_i + \theta_{iACE-BB},$$

if study i compares ACE inhibitors versus β-blockers or

$$\text{logit}(p_{iBB}) = u_i$$

$$\text{logit}(p_{iCCB}) = u_i + \theta_{iCCB-BB},$$

if study i compares CCB versus β-blockers.

Then the study-specific underlying treatment effects $\theta_{iACE-BB}$ and $\theta_{iCCB-BB}$ are distributed normally with expectations μ_{ACE-BB}, μ_{CCB-BB} and common heterogeneity τ^2 in the same way as the previous model. Using a half-normal prior distribution for the heterogeneity ($\tau \sim N(0, 1)$, $\tau > 0$) and fitting the model in WinBUGS produces the estimates presented in the lower part of Table 1.

The estimates obtained from the two approaches are very similar. The major difference between the two approaches is in the estimation of heterogeneity. The subgroup meta-analysis fitted within a Bayesian setting with the binomial likelihood gives a posterior median of τ^2 equal to 0.02 with 95% CrI (0.001,0.12), slightly larger than the point estimate from the frequentist meta-regression.

Indirect and Mixed Comparison

Theory and Formulae for Indirect Comparisons

In evidence-based medicine, estimates of treatment effect obtained from head-to-head RCTs and combined in a meta-analysis are widely considered the "best available" evidence with which

Table 1 Results of subgroup analysis for ACE inhibitors versus β-blockers and CCB versus β-blockers. Log-odds ratios (β) with their standard error $SE(\beta)$ and odds ratios (OR) with their 95% confidence or credible interval (CI/CrI) estimated from meta-regression and hierarchical models are reported

Model	Comparison	μ	SE (μ)	OR	95% CI/CrI for OR
Linear model in frequentist implementation	ACE inhibitors versus β-blockers	−0.17	0.10	0.84	(0.69,1.03)
	CCB versus β-blockers	−0.21	0.07	0.81	(0.71,0.93)
Hierarchical model in Bayesian implementation	ACE inhibitors versus β-blockers	−0.18	0.12	0.84	(0.66,1.06)
	CCB versus β-blockers	−0.21	0.09	0.81	(0.68,0.97)

to evaluate the effectiveness of medical interventions (Guyatt et al. 1995; McAlister et al. 1999). Consider two treatments, labelled B and C, which have been compared directly in RCTs and combined in a pairwise meta-analysis. The summary treatment effect estimate is denoted as $\hat{\mu}^D_{BC}$, where the superscript denotes the "direct" estimate and the subscript denotes the treatment comparison, where BC is the effect of C relative to B. In the absence of the "level one" evidence for B versus C, it has been suggested that an indirect estimate can be formed via a "common comparator" (Bucher et al. 1997; Song et al. 2003; Glenny et al. 2005), which is assumed to be treatment A. An indirect estimate $\hat{\mu}^I_{BC}$ can be derived by combining the meta-analytic effect estimates of A versus B studies $\left(\hat{\mu}^D_{AB}\right)$ and A versus C studies $\left(\hat{\mu}^D_{AC}\right)$, such that,

$$\hat{\mu}^I_{BC} = \hat{\mu}^D_{AC} - \hat{\mu}^D_{AB}$$

This method is often referred to as an "adjusted indirect comparison," so-called because randomization is respected by using the relative effect estimates $\hat{\mu}^D_{AC}, \hat{\mu}^D_{AB}$ obtained from the meta-analyses. Here it is referred to simply as an indirect comparison.

The usual measures of statistical variability can be derived for the indirect estimate. As $\hat{\mu}^I_{BC}$ is formed as a difference between two independent estimates its variance, $\hat{v}^I_{BC}$, is equal to the sum of the variances, $\hat{v}^D_{AC}$ and $\hat{v}^D_{AB}$, estimated from the direct AC and AB comparisons:

$$\hat{v}^I_{BC} = \hat{v}^D_{AC} + \hat{v}^D_{AB}$$

A single head-to-head randomized trial is as precise as an indirect comparison based on four trials of the same size. To see this, suppose each trial produces an estimate with variance σ^2. A meta-analysis of s trials with direct estimates of A versus B will have variance $\hat{v}^D_{AB} = \sigma^2/\mathrm{s}$ (based on inverse variance weights). The indirect estimate of B versus C via A based on s AB and s AC trials will have variance $\hat{v}^I_{BC} = \hat{v}^D_{AC} + \hat{v}^D_{AB} = \sigma^2/\mathrm{s} + \sigma^2/\mathrm{s} = 2\sigma^2/\mathrm{s}$.

A common misconception is that for an indirect comparison to be valid, every trial must include a common comparator (Hughes 2010). In truth, indirect estimates can be derived via many routes. The only requirement is that the network is "connected" and not necessarily via a common comparator. Consider the network shown in Fig. 2 which is adapted from a 2006 submission to NICE which included four distinct regimens for the second-line treatment of metastatic breast cancer (Eli Lilly 2006).

Table 2 reports the results for difference in median years survival. Note there are direct estimates available for gemcitabine + paclitaxel versus paclitaxel ($\hat{\mu}^D_{AB} = -2.8$ years), paclitaxel versus docetaxel ($\hat{\mu}^D_{BC} = 2.7$ years and docetaxel versus capecitabine + docetaxel ($\hat{\mu}^D_{CD} = 3.0$ years), which forms a "chain" of evidence A-B-C-D. The comparison of interest to the decision-maker was gemcitabine + paclitaxel (A) versus capecitabine + docetaxel (D) (Jones et al. 2006),

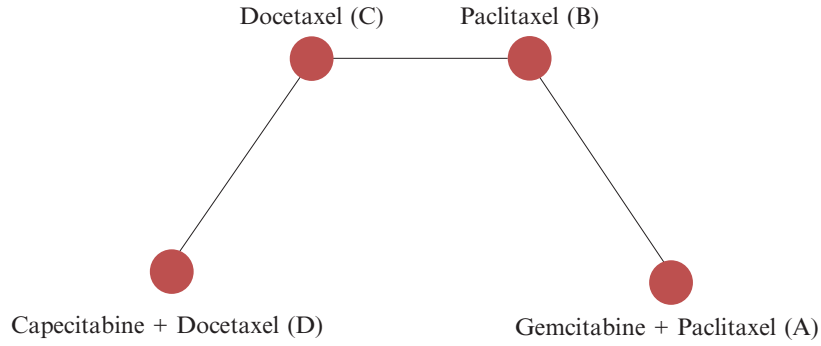


Fig. 2 Chain of comparisons network of chemotherapy treatments for second-line treatment of breast cancer

Table 2 Findings from the manufacturer's submission for gemcitabine STA. Median difference in survival and 95% confidence intervals (Adapted from: Eli Lilly 2006 and Jones et al. 2006)

Treatment comparison	Trials	Median difference (MD) (95% CI)	SE (MD)	Variance (MD)
Gemcitabine + paclitaxel (A) versus paclitaxel (B)	1	−2.8 (−0.01, −5.6)	1.42	2.02
Paclitaxel (B) versus docetaxel(C)	1	2.7 (0.3, 5.1)	1.24	1.54
Docetaxel (C) versus Capecitabine + docetaxel (D)	1	3.0 (0.6, 5.4)	1.20	1.44

an indirect comparison of which can be formed as

$$\hat{\mu}^I_{AD} = \hat{\mu}^D_{AB} + \hat{\mu}^D_{BC} + \hat{\mu}^D_{CD} = 2.9 \text{ years}$$

$$SE\left(\hat{\mu}^I_{AD}\right) = \sqrt{\hat{v}_{AB} + \hat{v}_{BC} + \hat{v}_{CD}} = 2.23$$

Theory and Formulae for Mixed Comparisons

If both direct and indirect estimates are available for the same comparison, they can be combined by taking the weighted average of μ^D_{BC} and $\hat{\mu}^I_{BC}$. This has been referred to as a *mixed* comparison and will be denoted here as μ^M_{BC}. However, it should not be confused with a "mixed-treatment comparison" (Lu 2004), which refers to the simultaneous comparison of multiple treatments in a single analysis and is synonymous to network meta-analysis. A simple and intuitive approach for combining direct and indirect evidence is the inverse variance method, where

$$\hat{\mu}^M_{BC} = \frac{\frac{1}{\hat{v}^D_{BC}}\hat{\mu}^D_{AB} + \frac{1}{\hat{v}^I_{BC}}\hat{\mu}^I_{BC}}{\frac{1}{\hat{v}^D_{BC}} + \frac{1}{\hat{v}^I_{BC}}}$$

with variance

$$\hat{v}^M_{BC} = \frac{1}{\frac{1}{\hat{v}^D_{BC}} + \frac{1}{\hat{v}^I_{BC}}}$$

A 95% confidence interval for the mixed estimate can be obtained as $\hat{\mu}^M_{BC} \pm 1.96\sqrt{\hat{v}^M_{BC}}$. Note that in the case of a dichotomous outcome, where μ is the ln(OR) or ln(RR), mean effect size and confidence intervals for the OR and RR can obtained by exponentiation.

Example: Indirect and Mixed Comparison for ACE inhibitors Versus CCB

Using the ln(*OR*) for the comparisons ofACE inhibitors and CCB each versus β-blockers presented in Table 1, an indirect estimate for ACE inhibitors versus CCB can be obtained. The indirect ln(*OR*) estimate $\hat{\mu}^I_{ACE-CCB}$ is calculated as the difference between the direct ln(*OR*) for CCB versus β-blockers and direct ln(*OR*) for ACE inhibitors versus β-blockers. Using the estimates from the frequentist subgroup analysis described earlier, the indirect estimate is

$$\begin{aligned}\hat{\mu}^I_{ACE-CCB} &= \hat{\mu}^D_{ACE-BB} - \hat{\mu}^D_{CCB-BB} \\ &= -0.17 - (-0.21) = 0.04\end{aligned}$$

The variance of the indirect estimate $\hat{\mu}^I_{ACE-CCB}$ is the sum of the variances of $\hat{\mu}^I_{ACE-BB}$ and $\hat{\mu}^D_{CCB-BB}$:

$$\begin{aligned}\hat{v}^I_{ACE-CCB} &= \hat{v}^D_{ACE-BB} + \hat{v}^D_{CCB-BB} \\ &= 0.10^2 + 0.07^2 = 0.0149\end{aligned}$$

Therefore, the indirect OR of ACE inhibitors versus CCB is $\exp\left(\hat{\mu}^I_{ACE-CCB}\right) = 1.04$ with 95% CI $\exp\left(\hat{\mu}^I_{ACE-CCB} \pm 1.96\sqrt{\hat{v}^I_{ACE-CCB}}\right) = (0.82, 1.32)$.

Since there are also three studies that directly compare ACE inhibitors with CCB, they can be combined with the indirect estimate to produce a mixed estimate. Synthesis of the studies provides a direct estimate for the $\hat{\mu}^D_{ACE-CCB}$ equal to -0.22 with standard error 0.11. Then, the mixed estimate can be obtained as the weighted average of the direct and indirect ln(*OR*):

$$\hat{\mu}^M_{ACE-CCB} = \frac{\frac{1}{\hat{v}^D_{ACE-CCB}}\hat{\mu}^D_{ACE-CCB} + \frac{1}{\hat{v}^I_{ACE-CCB}}\hat{\mu}^I_{ACE-CCB}}{\frac{1}{\hat{v}^D_{ACE-CCB}} + \frac{1}{\hat{v}^I_{ACE-CCB}}}$$

$$= \frac{\frac{1}{0.0121}(-0.22) + \frac{1}{0.0149}0.04}{\frac{1}{0.0121} + \frac{1}{0.0149}} = -0.10$$

The variance of this estimate is

$$\hat{v}^M_{ACE-CCB} = \frac{1}{\frac{1}{0.0121} + \frac{1}{0.0149}} = 0.0067$$

$$= \frac{1}{\frac{1}{0.0225} + \frac{1}{0.0149}} = 0.009$$

The mixed OR is $\exp\left(\hat{\mu}^M_{ACE-CCB}\right) = 0.90$ with 95% CI (0.77, 1.06).

The indirect and mixed ORs for ACE inhibitors versus CCB are presented in Fig. 3. Note that the confidence interval for the mixed estimate is narrower than the confidence interval for the direct estimate. By combining direct evidence with indirect evidence, the variance is reduced by 45%.

This approach is intuitive and will be familiar to meta-analysts; however, it is labor-intensive. In a three treatment network, A-B-C, the meta-analytic estimates $\hat{\mu}^D_{AB}$, $\hat{\mu}^D_{AC}$, $\hat{\mu}^D_{BC}$ must be obtained, the indirect estimates $\hat{\mu}^I_{AB}$, $\hat{\mu}^I_{AC}$, $\hat{\mu}^I_{BC}$ derived, and then the mixed estimates $\hat{\mu}^M_{AB}$, $\hat{\mu}^M_{AC}$, $\hat{\mu}^M_{BC}$ computed. As the number of treatments increases and the network expands, this approach quickly becomes untenable and more sophisticated approaches can be used (Caldwell et al. 2005). Section "Models for Network Meta-analysis" discusses methods for simultaneously combining direct and indirect evidence in a single analysis. The next section discusses the underlying assumptions needed to undertake an indirect or mixed comparison.

Assumptions Underlying Indirect and Mixed Comparisons

Current hierarchies of evidence place indirect and "mixed" comparisons below direct evidence regardless of whether the constituent effect estimates have been obtained from meta-analyses of RCTs (currently "level one" evidence). Several HTA organizations have expressed doubts about indirect comparisons and state that if direct evidence exists, it should take precedence. For example, the *Cochrane Handbook* (Higgins and Green

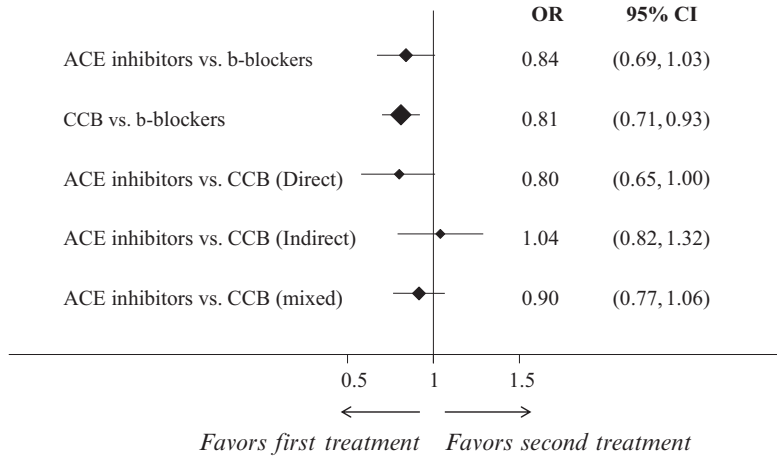


Fig. 3 Summary odds ratios (*OR*) for each comparison using direct, indirect, and mixed evidences. Diamonds represent the point estimates and the horizontal lines the corresponding 95% confidence intervals

2008) states "indirect comparisons may suffer the biases of observational studies" and advises that direct and indirect evidence only be combined as a supplemental analysis. In England and Wales, NICE (2008) uses direct evidence as the reference case for appraisals of new technologies, only allowing indirect and "mixed" comparisons as a supporting analysis. Similarly, CADTH (Wells et al. 2009) adopts a cautious stance and the Pharmaceutical Benefits Advisory Committee (PBAC) in Australia have expressed skepticism about the use of indirect evidence (PBAC 2008).

This caution is based on concerns regarding the key assumption underpinning indirect comparisons, which is reflected mathematically in the *consistency equation* $\mu_{BC} = \mu_{AC} - \mu_{AB}$. The consistency equation relates to the true (or average) effectiveness of B versus C rather than to each individual study. It states that the effect of B versus C can be estimated either indirectly via A (right part of the equation) or directly (left part of the equation) and that these two pieces of evidence will, on average, give the same result. Rearranging the parts of the equation shows that one consistency equation is sufficient to reflect consistency for all three comparisons in a three treatment network. Such that

$$\mu_{BC} = \mu_{AC} - \mu_{AB}, \mu_{AB} = \mu_{AC} - \mu_{BC},$$
$$\mu_{BC} = \mu_{AC} - \mu_{AB}$$

The validity of the consistency equation is fundamental to the validity of indirect comparisons. In considering the validity of the assumption for the combination of direct and indirect evidence, some authors have found it instructive to separate the notion of *similarity* (Song et al. 2009; Donegan et al. 2010) or *transitivity* (Baker and Kramer 2002) from the notion of *consistency*. In the current chapter, these notions are interpreted as the distinction between clinical or epidemiological considerations on the one hand, and statistical considerations on the other. Transitivity refers to the genuine ability to learn about a pairwise comparison via an intermediate treatment via indirect comparison. As will be discussed below, it requires the intermediate treatment to be equivalent when compared against each of the treatments of interest and that the actual studies contributing to the indirect comparison do not differ in important ways. Specifically, when μ^{I}_{AB} is calculated, it is assumed that we can learn about B versus C via A. The common comparator A might be said to be "transitive" when it allows valid comparison of the treatments to which it is linked. Note that transitivity is not a property of the common treatment A alone but of the two sets of studies it links.

Consistency is a statistical notion that can be considered at the level of the parameters or the level of the data. The consistency equation defines relationships among the parameters. The validity

of the assumption embodied in the equation can be assessed only when data from different sources form a "closed loop" of evidence in the network (a path that starts and ends at the same node treatment). When the consistency assumption does not hold or when there is evidence of disagreement between direct and indirect evidence, then the evidence is said to be *inconsistent* (or show *inconsistency*).

When transitivity holds and there are multiple sources of evidence, the consistency equation should hold. The consistency equation may hold in a statistical sense, however, even when the studies do not allow valid indirect comparisons, due to important differences between studies that prevent an assumption of transitivity from holding. If there is inconsistency in the data, the assumption of transitivity is clearly challenged. For an "open loop," that is one for which there is indirect evidence but no direct evidence, consistency cannot be evaluated statistically, and the validity of the indirect comparison must rest entirely on clinical and epidemiological judgements regarding the plausibility of transitivity. It can be shown mathematically (Lu and Ades 2009) that consistency is a consequence of the assumption of exchangeability that forms the basis of the Bayesian network meta-analysis models which is, in turn, an extension of the usual assumption made in a pairwise meta-analysis (Dias et al. 2010). The assumption of transitivity is essentially equivalent to the assumption of exchangeability in this sense, since it relates to similarity of studies. The term "transitivity" might be preferred to "similarity" (Donegan et al. 2010); however, because (i) it better describes the aim of the assumption to compare two treatments via a third one; (ii) it clearly refers to more than two comparisons whereas the term "similarity" reduces to homogeneity when we refer to a single head-to-head comparison; and (iii) "similarity" may be misinterpreted as necessitating *all* trial and patient characteristics to be similar, when in truth a valid indirect comparison can be obtained even when studies are dissimilar, so long as such characteristics do not modify treatment effect.

Requirements for Transitivity

Transitivity requires some particular characteristics of the studies contributing to the indirect comparison, as follows:

- The two sets of trials AB and AC do not differ with respect to the distribution of effect modifiers.

In order for an indirect comparison to be valid, the distribution of treatment effect modifiers should be similar in AB and AC trials. Before conducting an indirect or mixed comparison, the analyst should therefore ensure they have identified a priori possible effect modifiers and should compare their distributions across treatment comparisons. For example, in a network of treatments for childhood nocturnal enuresis, Caldwell et al. (2010) hypothesize that the age of the children could be a potential effect modifier since a 6-year-old suffering from nighttime bedwetting might have a different underlying pathology from a 12-year-old.

Note that the consistency assumption holds at the level of the mean effect sizes and as such an effect modifier that differs within studies of one comparison but has a similar distribution across both comparisons will not violate the assumption. For example, if age is an effect modifier and AC trials differ in terms of mean age of participants (which will manifest as heterogeneity in AC studies), but the same variability is observed in the set of AB trials then transitivity could still hold. In contrast, if the distribution of age differs across comparisons such that children in the AC studies tended to be younger and those in AB studies tended to be much older, then the assumption would not hold.

Adjustment using regression techniques can be used to account for small differences in the distribution of effect but note that covariates must be carefully selected. Only effect modifiers and not "colliders" (variables that influence the choice of comparison and the effectiveness) should be considered for adjustment. Adjustment for colliders, as in classical epidemiology, will introduce bias

rather than improve the plausibility of the transitivity assumption (Jansen et al. 2012).

- The interventions are being given for the same indications.

Transitivity could be violated if interventions have different indications, so it is important that sets of studies contributing to the indirect comparison are using the treatments for the same underlying condition. A particularly useful way to think about this requirement is to consider whether the participants included in the network could, in principle, be randomized to any of the three treatments A, B, and C. For example, if treatment A is a chemotherapy regimen typically administered as a second-line treatment, whereas treatments B and C can be used either as first or second line, it cannot be assumed that participants in a BC trial could have been randomized in an AC trial. Although this consideration is a fundamental one and should be addressed when building the evidence network, it might be the case that treatments are comparable in theory but not in practice. For example, interferon, glatiramer acetate, or natalizumab are commonly used in clinical practice for patients with relapsing-remitting multiple scleroses whereas mitoxantrone, methotrexate, cyclophosphamide, or azathioprine are more frequent for patients with a progressive disease. Evidence to support this clinical "tradition" is not solid, and it would be appealing to compare all these treatments. In practice, however, transitivity will be violated as comparisons will differ in disease severity.

Another way to conceptualize this requirement is to consider treatments not included in each study as missing data (Lu and Ades 2006). Thus, AB trials are missing C arms, and AC trials are missing B arms. The transitivity assumption is likely to hold if these arms are missing in an entirely random way, which guarantees that the choice of treatments is unrelated to the indications for which they are given. In practice, the selection of treatments to be included in a trial is not random. In many clinical trials, the choice of comparator is placebo or an older, suboptimal intervention rather than a realistic alternative such as an established effective treatment. If the choice of comparator is associated, directly or indirectly, with the relative effectiveness of the interventions then the key assumption will be violated.

- The treatment C is similar when it appears in AC and BC trials.

The transitivity assumption is violated when the treatments in question differ *systematically* between trials. The definition of the nodes in the treatment network is a challenging issue as very often treatments are given at various doses, administrations routes, frequencies, etc. For example, consider that the common comparator A is a treatment which can be given at different doses, but there is no systematic difference on the average dose of A between AC and AB trials. In this case the assumption can hold although there could be heterogeneity within AC and AB comparisons. Consequently, the "anchor" treatment A can be represented by a single node allowing the indirect comparison of B and C. If, however, A is given via a different administration route in all AC and AB trials, then it is questionable whether the two types of A can form a common node and an indirect comparison of B versus C via A would be impossible. For example, when comparing different fluoride treatments, comparison between fluoride toothpaste and fluoride rinse can be made via placebo. However, placebo toothpaste and placebo rinse might not be comparable as the mechanical function of brushing might have a different effect on the prevention of caries. If this is the case, the transitivity assumption is doubtful (Salanti et al. 2009).

Estimating Inconsistency in Mixed Comparisons

In theory, the consistency equation $\mu_{BC} = \mu_{AC} - \mu_{AB}$ must hold if transitivity is deemed to hold. However, in practice, there may be inconsistency in the evidence base. In a three-treatment network, three independent direct estimates, $\hat{\mu}^D_{AB}$, $\hat{\mu}^D_{AC}$ and $\hat{\mu}^D_{BC}$ (assuming there are no trials with more than

two arms), and three indirect estimates, $\hat{\mu}^I_{AB}$, $\hat{\mu}^I_{AC}$ and $\hat{\mu}^I_{BC}$, can be obtained. Assuming the treatment comparison of interest is B versus C, the discrepancy (difference) between the direct and indirect estimates forms the measure of inconsistency. This discrepancy is called the inconsistency factor (IF) which is estimated as

$$\widehat{IF}_{ABC} = \left|\hat{\mu}^D_{BC} - \hat{\mu}^{\mathrm{I}}_{BC}\right|$$

Note that the direction of the difference might be clinically important but mathematically is unimportant for the statistical evaluation of consistency. Consequently only absolute differences are taken. In a three-treatment network, only one measure of inconsistency is possible (and hence the subscript denoting the loop) (Lu and Ades 2006) as it can be shown that the same inconsistency factor will be obtained whichever edge of the triangle is of interest.

The variance of the inconsistency factor is

$$\mathrm{var}\left(\widehat{IF}_{ABC}\right) = \hat{v}^D_{BC} + \hat{v}^{\mathrm{I}}_{BC}$$

A 95% confidence interval can be obtained for the inconsistency factor as $\widehat{IF}_{ABC} \pm 1.96 \sqrt{\mathrm{var}\left(\widehat{IF}_{ABC}\right)}$. The null hypothesis of evidence consistency $\widehat{IF} = 0$ can then be tested by deriving a z-test (Bucher et al. 1997).

$$z = \frac{\widehat{IF}_{ABC}}{\sqrt{\mathrm{var}\left(\widehat{IF}_{ABC}\right)}}$$

If consistency holds, it is reasonable to combine across $\hat{\mu}^D_{BC}$ and $\hat{\mu}^I_{BC}$ to form $\hat{\mu}^M_{BC}$. However, if there is evidence of a "statistically significant" discrepancy ($p \leq 0.05$), the fundamental assumption is not fulfilled, and one may say that there is evidence of inconsistency.

Claims have been made that indirect comparisons may systematically over- (Bucher et al. 1997; Mills et al. 2011) or underestimate treatment effects compared with direct comparisons. Since inconsistency is a property of a "loop" of evidence apparent overestimation of a treatment effect on one side of a triangle network (e.g., $\hat{\mu}^I_{BC}$) corresponds to underestimation of another (e.g., $\hat{\mu}^I_{AC}$). Thus, any assessment of consistency needs to take account of the particular circumstances of the problem. Until recently, empirical investigation of the extent of inconsistency has been limited. In a recent review, (Song et al. 2011) examined 112 independent three-treatment networks and detected 16 cases of statistically significant discrepancies between direct and indirect estimates. However, there was no consistent direction as to over- or underestimation. Of course, the test for inconsistency may have low power to detect true inconsistency should it exist, as with other interaction effects. The analyst must therefore be extremely cautious in their interpretation even if inconsistency is not detected.

Note that the discovery of inconsistency does not necessarily mean that all indirect comparisons in the loop are invalid. For example, suppose that AC and BC trials are similar regarding the distribution of effect modifiers (e.g., all studies are carried out in adults with a similar distribution in age), so that $\hat{\mu}^I_{AB}$ is a valid estimate of the relative effectiveness of A versus B for the given setting and population. If now the AB studies have all being carried out in younger populations (e.g., in adolescents) then the consistency assumption does not hold; both $\hat{\mu}^I_{AB}$ and $\hat{\mu}^D_{AB}$ are valid but answer different questions; hence computation of a mixed estimate, $\hat{\mu}^M_{AB}$, would be inappropriate.

Example: Inconsistency in the Evidence Triangle ACE Inhibitors Versus CCB Versus β-Blockers

Inconsistency can be evaluated by calculating the difference between direct and indirect estimate for the same comparison. In the case of ACE inhibitors versus CCB, the inconsistency factor reflects the disagreement in the triangle formed by the three sets of trials ACE inhibitors versus CCB versus β-blockers and is calculated as

$$\widehat{IF}_{ACE-CCB-BB} = \left|\hat{\mu}^D_{ACE-CCB} - \hat{\mu}^I_{ACE-CCB}\right| = |-0.22 - 0.04| = 0.26$$

The standard error of the inconsistency factor is obtained as

$$SE\left(\widehat{IF}_{ACE-CCB-BB}\right) = \sqrt{\hat{v}^D_{ACE-CCB} + \hat{v}^I_{ACE-CCB}}$$
$$= \sqrt{0.0144 + 0.0225}$$
$$= 0.192$$

A 95% CI for the inconsistency factor is obtained as $\widehat{IF} \pm 1.96 \times SE\left(\widehat{IF}\right) = (-0.12, 0.64)$. The z-test for the hypothesis $H_0 : \hat{IF}_{ACE-CCB-BB} = 0$ is

$$z = \frac{\widehat{IF}_{ACE-CCB-BB}}{SE\left(\widehat{IF}_{ACE-CCB-BB}\right)} = \frac{0.26}{0.192} = 1.35$$

leading to a p-value equal to 0.91. Note that this result applies to the entire triangle: the same inconsistency factor and p-value could have been obtained by calculating the difference between direct and indirect evidence for the ACE versus β-blockers or CCB versus β-blockers comparisons. As the 95% CI includes zero, there is no indication of important statistical inconsistency between direct and indirect estimate, which is also supported by the p-value.

Models for Network Meta-analysis

Extensions of the ideas above to more than three treatments lead to a general framework for network meta-analysis. Consider a set of T treatments of interest that we want to evaluate according to their relative effectiveness on a single outcome measure. The treatments are studied collectively in N studies. Each study may provide evidence about some of the treatments; it will include only a subset of T, $T_i \in T$. The study data can be arm-based or contrast-based. In the contrast-based approach, the effect sizes y_{ijk} from each study are available, and they refer to the relative effectiveness of a treatment k relative to j with $j, k \in T_i$ Network meta-analysis can be viewed as a special case of meta-regression (linear model), as a hierarchical model or as a multivariate meta-analysis model. The estimation methods that arise from these approaches are essentially equivalent and can be employed under the assumption of consistency or under assumptions that impose fewer restrictions.

A key issue in all methods for fitting network meta-analysis is the minimization of the parameters' space by selecting a minimum set of *basic parameters.* This is a set of comparisons (as many as the total number of treatments minus one) that are sufficient to generate all possible comparisons between the treatments via the consistency equations. Under consistency, the choice of the basic parameters does not affect the results but typically the basic parameters are defined by taking the comparisons of all treatments versus a common reference to simplify interpretation. Examples to follow should make this clear.

Consistency Models

Network Meta-analysis as a Linear Model

Consider the simplest case of having three treatments of interest $T = \{A, B, C\}$ and studies that compare all possible pairs of those treatments; i.e., there are AB, AC, and BC studies. For now it is assumed that only two-arm trials are available. In general, y_{ijk} refers to the relative effectiveness of two interventions j and k within study i.When each study has only two treatments, the treatment indices can be dropped and the observed effect written as y_i. The treatment indices will be reemployed in section "Network Meta-analysis as a Hierarchical Model."

The two-step process described in sections "Theory and Formulae for Indirect Comparisons" and "Theory and Formulae for Mixed Comparisons" is a simple network meta-analysis. For a given comparison, say B versus C, an indirect estimate $\hat{\mu}^I_{BC}$ is derived by combining AC and AB studies. Then, the indirect estimate and direct estimate $\hat{\mu}^D_{BC}$ are synthesized to obtain the mixed summary estimate $\hat{\mu}^M_{BC}$. This first step of this process was described in the context of a meta-regression in section "Meta-analysis and Meta-regression as Linear Model." This can be done by creating two dummy variables to identify the AC and BC studies and omitting the intercept:

$$y_i = \mu_{AC} x_{iAC} + \mu_{AB} x_{iAB} + \delta_i + e_i$$

The same model can be used for both stages of the mixed comparison analysis, by careful specification of the covariate values in a way that forces the consistency equation into the analysis as a constraint. As above, if study i compares A and C, then $x_{iAC} = 1$, $x_{iAB} = 0$, and if study i compares A and B then $x_{iAC} = 0$, $x_{iAB} = 1$. Now if study i compares B and C, the consistency equation $\mu_{BC} = \mu_{AC} - \mu_{AB}$ can be introduced by setting $x_{iAC} = 1$, $x_{iAB} = -1$. Note that because of the assumption of consistency, only two comparisons need to be included in the model (here AC and AB), and consequently there are two explanatory variables to be included in the meta-regression model. The two contrasts AC and AB are called the *basic contrasts* and the parameters μ_{AC}, μ_{AB} the *basic parameters,* whereas the μ_{BC} is a functional parameter and can be derived as a linear function of the two basic parameters. The choice of the two out of the three contrasts that enter the meta-regression is arbitrary and does not impact on the parameters estimation; e.g., x_{iAC} and x_{iBC} could have been chosen as covariates.

Extending the idea to the case of more than three treatments results in a full network meta-analysis. For example, with $T = \{A, B, C, D, E\}$ treatments included, there are $T(T-1)/2 = 10$ possible head-to-head comparisons. $T - 1$ basic parameters are selected, such as all treatment comparisons Aj of treatment $j = B$, C , D , E versus treatment A, relating to regression coefficients μ_{Aj}. The meta-regression model would be

$$y_i = \sum_{j=B,C,D,F} \mu_{Aj} x_{iAj} + \delta_i + e_i \qquad (6)$$

with $e_i \sim N\left(0, s_i^2\right)$. The variable $x_{iAj} = 1$ if study i compares A and j, $x_{iAj} = 0$ if study i compares A and k , $k \neq j$. If a study compares treatment s, j and k and does not include A, then the consistency equations are used to derive the values of x_{iAj}. If, for example, a study compares B and D, then $x_{iAj} = 0, j = C$, E, $x_{iAB} = -1$ and $x_{iAD} = 1$ because $\mu_{BD} = \mu_{AD} - \mu_{AB}$.

In summary, all observed comparisons are reexpressed using the regression covariates x_{iAj}. This gives the model $N_{\text{comp}} - (T-1)$ degrees of freedom (number of functional parameters), where N_{comp} is the number of comparisons observed in the network. For example, if the network consists of AC , AB , BC , CD , BE , BD studies, then there are $6 - (5-1) = 2$ degrees of freedom. This can be also visualised by the number of independent closed loops in the network diagram (ABC and BCD).

This model can be fitted using any meta-regression software (such as the metareg command in STATA). The estimated regression coefficients $\hat{\mu}_{Aj}$ are network meta-analysis estimates for all treatments versus the reference treatment A and their uncertainty is conveyed by $SE\left(\hat{\mu}_{Aj}\right)$. Network meta-analysis summary effects for all other comparisons, say B versus D, can be obtained by considering the consistency equations relating the $\hat{\mu}_{Aj}$ to the functional parameters. Their variances can be obtained by combining standard errors and covariances (from the variance-covariance matrix for the estimated regression coefficients). For instance, $\hat{\mu}_{BD} = \hat{\mu}_{AD} - \hat{\mu}_{AB}$ and $SE^2(\hat{\mu}_{BD}) = SE^2(\hat{\mu}_{AD}) + SE^2(\hat{\mu}_{AB}) - 2\text{cov}(\hat{\mu}_{AD}, \hat{\mu}_{AB})$.

Note that the random effects follow a normal distribution $\delta_i \sim N(0, \tau^2)$, with heterogeneity variance assumed to be equal for every comparison. This may be a strong assumption as different comparisons might include studies with different between-study variability. Assuming a common heterogeneity might impose an inappropriate τ^2 value for some comparisons. Although assuming comparison-specific heterogeneities can be desirable in many cases, it presents practical difficulties. Estimation of the parameter τ^2 can be challenging if few studies are available. Even with large network meta-analyses including many treatments, it is often the case that some of the comparisons include only a few studies; some comparisons might even be informed by a single study. Nevertheless, assuming a common heterogeneity parameter allows comparisons to "borrow strength" from each other in the estimation of the common τ^2, overcoming computational problems

that are encountered both with frequentist and Bayesian fitting of models.

Application: Network Meta-analysis Using Meta-regression for Incident Diabetes

Standard meta-regression methods can be only be applied to networks that contain two-arm studies. The following analysis treats the 30 pairwise comparisons in the incident diabetes data set as if they came from 30 (rather than the true 22) independent studies. A meta-regression model is be employed where again the different comparisons define the subgroups. First the $T - 1$ "basic contrasts" need to be selected, to be included as covariates in the model. Several combinations of basic contrasts are possible, and for $T = 6$, five parameters need to be selected. For ease of interpretation, it is convenient to choose the comparisons of each treatment versus a common reference treatment. Here, placebo (P) is chosen to be the reference treatment and basic contrasts are defined for each treatment versus placebo. Then, to specify the design matrix all comparisons in the network need to be written as functions of the basic parameters. The first two columns of Table 3 list all comparisons in the network for which direct estimates are available and the number of studies involving each comparison. Then, for the five comparisons belonging to the basic contrasts (e.g., β-blockers (BB) vs. P), the respective variable x_i (x_{iBB-P}) takes the value 1 and the variables of the other four basic contrasts take the value 0. For any other treatment comparison (e.g., diuretics (D) vs. BB) x_i takes value -1 for the first treatment (x_{iD-P}) and -1 for the second treatment based on the consistency equations ($\mu_{D-BB} = \mu_{D-P} - \mu_{BB-P}$).

Table 3 Parameterization of design matrix for the five basic contrasts when placebo is the reference treatment for incident diabetes

Comparison in study ***i***	Number of studies	x_{iBB-P} β-blockers versus placebo	x_{iD-P} diuretics versus placebo	x_{iCCB-P} CCB versus placebo	x_{iACE-P} ACE inhibitors versus placebo	x_{iARB-P} ARB versus placebo
β-blockers versus placebo	1	1	0	0	0	0
diuretics versus placebo	3	0	1	0	0	0
diuretics versus β-blockers	2	−1	1	0	0	0
CCB versus placebo	1	0	0	1	0	0
CCB versus β–blockers	5	−1	0	1	0	0
CCB versus diuretics	2	0	−1	1	0	0
ACE inhibitors versus placebo	2	0	0	0	1	0
ACE inhibitors versus β–blockers	3	−1	0	0	1	0
ACE inhibitors versus CCB	2	0	0	−1	1	0
ARB versus placebo	3	0	0	0	0	1
ARB versus β–blockers	3	−1	0	0	0	1
ARB versus diuretics	1	0	−1	0	0	1
ARB versus CCB	1	0	0	−1	0	1

Table 4 Results of network meta-analysis as meta-regression for incident diabetes. Log-odds ratios ($\hat{\mu}$) with their standard error $SE(\hat{\mu})$ and odds ratios (OR) with their 95% confidence interval (CI) for all basic and functional contrasts are reported

Comparison	$\hat{\mu}$	$SE(\hat{\mu})$	OR	95% CI for OR
β-blockers versus placebo	0.24	0.09	1.27	(1.07, 1.52)
diuretics versus placebo	0.32	0.09	1.38	(1.15, 1.64)
CCB versus placebo	0.08	0.08	1.08	(0.93, 1.27)
ACE inhibitors versus placebo	−0.11	0.08	0.90	(0.77, 1.05)
ARB versus placebo	−0.17	0.10	0.84	(0.69, 1.03)

Table 5 Variance-covariance matrix of the five basic parameters used in meta-regression approach for network meta-analysis for incident diabetes. The diagonal includes the variances of the parameters and the other cells the covariances between the two corresponding parameters

	$\hat{\mu}_{BB-P}$	$\hat{\mu}_{D-P}$	$\hat{\mu}_{CCB-P}$	$\hat{\mu}_{ACE-P}$	$\hat{\mu}_{ARB-P}$
$\hat{\mu}_{BB-P}$	0.0076				
$\hat{\mu}_{D-P}$	0.0040	0.0080			
$\hat{\mu}_{CCB-P}$	0.0052	0.0040	0.0070		
$\hat{\mu}_{ACE-P}$	0.0038	0.0034	0.0035	0.0058	
$\hat{\mu}_{ARB-P}$	0.0037	0.0024	0.0037	0.0022	0.0098

The full meta-regression model is

$$y_i = \mu_{BB-P}x_{iBB-P} + \mu_{D-P}x_{iD-P} + \mu_{CCB-P}x_{iCCB-P} + \mu_{ACE-P}x_{iACE-P} + \mu_{ARB-P}x_{iARB-P} + \delta_i + e_i.$$

Fitting the model in STATA using **metareg** produces the regression coefficients in Table 4.

The common heterogeneity parameter of the network was estimated as 0.02. The variance-covariance matrix of the regression-coefficients is saved by STATA as the "e(v)" matrix and can be obtained after fitting the meta-regression model (Table 5).

Then any head-to-head comparison can be derived applying again the consistency equations to the point estimates. For example, the ln(OR) of diuretics versus β-blockers is $\hat{\mu}_{D-BB} = \hat{\mu}_{D-P} - \hat{\mu}_{BB-P} = 0.32 - 0.24 = 0.08$, and its standard error is

$$SE(\hat{\mu}_{D-BB}) = \sqrt{\hat{v}_{D-P} + \hat{v}_{BB-P} - 2\mathrm{Cov}(\hat{\mu}_{D-P}, \hat{\mu}_{BB-P})}$$

$$= \sqrt{0.008 + 0.0076 - 2 \times 0.004} = 0.09$$

All other functional contrasts estimates are derived the same way. The network meta-analysis estimates for all comparisons are presented in the black diamonds in Fig. 4.

Note that the confidence intervals for the comparison ACE inhibitors versus CCB have further reduced compared with the direct or mixed estimate previously calculated. Also, an estimate for the comparison ARB versus ACE inhibitors is obtained for which no studies exist. Figure 4 also shows the ranking of the treatments by ranking the mean OR of each treatment versus placebo.

Network Meta-analysis as a Hierarchical Model

An alternative way to fit the network meta-analysis model is by extending the hierarchical model. For the simplest case of three treatments A, B, C, assume there are studies that inform all possible comparisons. The effect size for a study that compares A versus B is denoted by y_{iAB}. When only two-arm trials are included in the network, the likelihood for the observations is specific to the comparison being presented, i.e.,

$$y_{iAC} \sim N\left(\theta_{iAC}, s^2_{iAC}\right), y_{iAB} \sim N\left(\theta_{iAB}, s^2_{iAB}\right), y_{iBC} \sim N\left(\theta_{iBC}, s^2_{iBC}\right),$$

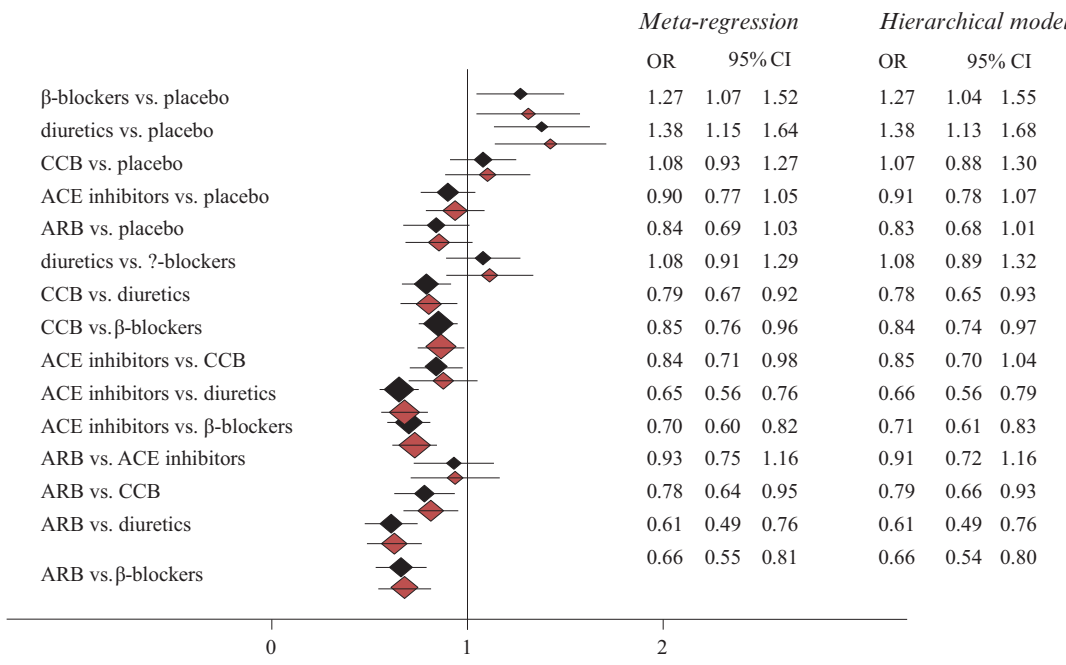


Fig. 4 Results from network meta-analysis conducted as meta-regression in STATA (*black*) ignoring correlations from multi-arm trials and as hierarchical model in WinBUGS that account for correlations (*red*). Diamonds are the point estimates of summary odds ratios (*OR*) and the horizontal lines represent the corresponding 95% confidence intervals (*CI*)

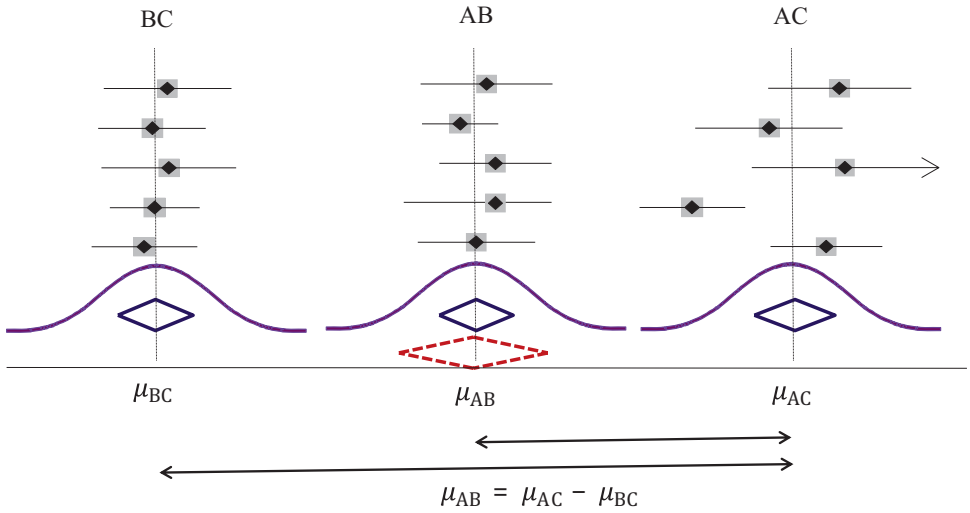


Fig. 5 Hypothetical example from three sets of meta-analyses that form a closed loop of evidence. The diamonds represent the summary effects using a random effects meta-analysis and a common heterogeneity parameter. The diamond in dashed line is the indirect estimate for the AB comparison

and similarly for the random effects

$$\theta_{iAC} \sim N(\mu_{AC}, \tau^2), \theta_{iAB} \sim N(\mu_{AB}, \tau^2),$$
$$\theta_{iBC} \sim N(\mu_{BC}, \tau^2).$$

This situation is depicted in Fig. 5.

The model so far is a collection of three independent meta-analyses. The three distributions relate to three different means, one mean per comparison. The consistency assumption claims that the three means are related via $\mu_{BC} = \mu_{AC} - \mu_{AB}$. This constraint results in the indirect estimation of B versus C and, if there are studies making the comparison directly, results also in the synthesis of the indirect evidence with the direct evidence. Indirect estimation is represented by the dashed diamond in Fig. 5. Of course the consistency relationship works by estimating indirect and mixed estimates for all comparisons, not just B versus C.

Extending the idea to more than three treatments is straightforward. For any two treatments $j, k = \{A, B, C, D, E\}$ compared in study i, the likelihood and the random effects distributions can be written as in a subgroup analysis, i.e., specific to the comparison j versus k

$$y_{ijk} \sim N\left(\theta_{ijk}, s^2_{ijk}\right) \tag{7}$$

$$\theta_{ijk} \sim N\left(\mu_{jk}, \tau^2\right) \tag{8}$$

Assuming consistency, the means of the random effects distributions are related. Selecting again $T - 1$ basic parameters μ_{Aj}, all means are related via $\mu_{jk} = \mu_{Ak} - \mu_{Aj}$. There are as many consistency equations as comparisons-specific meta-analysis, that is, N_{comp}, each equation expressing every comparison that appears in the data as a combination of the basic parameters. This gives the model $N_{\text{comp}} - (T - 1)$ degrees of freedom.

Note that the formulation above assumes that the different comparisons have the same heterogeneity variance, as in the meta-regression model in section "Network Meta-analysis as a Linear Model." The impact of this assumption can be better visualized in Fig. 5, where the three forest plots appear to reflect different degrees of heterogeneity. Nevertheless, the three random effects distributions have the same dispersion as a result of imposing a common heterogeneity parameter of their variance τ^2. This will "inflate" the uncertainty in the summary estimates in the more homogeneous sets of studies BC and AB and "deflate" the uncertainty in the AC estimate by attaching a lower τ^2 value for this comparison. A notable consequence is that the network meta-analysis summary effect size for a particular comparison can be less precise than the summary effect size from direct evidence alone. This can happen when a comparison with very low or no heterogeneity enters a network that consists of heterogeneous comparisons. Then, the estimated common heterogeneity parameter (which will be higher than the true for the homogeneous comparison) will impose greater uncertainty in the estimate for the homogeneous comparison, which might outweigh the gain in precision offered by including the indirect evidence.

Fitting the hierarchical model within a Bayesian framework makes the use of the true likelihood for the data easier. In the case of dichotomous outcomes, each study reports numbers of successes per arm and the likelihood (Eq. 7) is substituted by two arm-specific binomial distributions

$$r_{ij} \sim B\left(p_{ij}, n_{ij}\right)$$

$$r_{ik} \sim B\left(p_{ik}, n_{ik}\right)$$

Then the probabilities p_{ik}, p_{ij} are parameterized to produce a treatment effect measure θ_{ijk} (e.g., for log(OR), $\theta_{ijk} = \text{logit}(p_{ij}) - \text{logit}(p_{ik})$). The hierarchical network meta-analysis model is mathematically equivalent to the meta-regression as long as the contrast-specific data are used: they both have the same number of degrees of freedom and the same number of parameters.

Application: Network Meta-analysis as Hierarchical Model for Incident Diabetes

As in the meta-regression approach, all comparisons included in the four three-arm trials are

assumed to be evaluated in three independent two-arm studies, and this results in 30 comparisons indexed with i. The same five basic parameters μ_{Aj} are chosen, with placebo as reference: μ_{BB-P}, μ_{D-P}, μ_{CCB-P}, μ_{ACE-P}, μ_{ARB-P}. Arm-specific data will be modelled using the binomial likelihood. A categorical covariate needs to be specified for each arm, with x_{ij} showing the intervention given to arm j of study i and x_{ik} the intervention of arm k (x_{ij} , $x_{ik} = \{P, BB, D, CCB, ACE, ARB\}$). Fitting the model in WinBUGS, and using a half-normal prior distribution $\tau \sim N(0, 1)$, $\tau > 0$ for the common heterogeneity, gives the results in Table 6.

These estimates are comparable to the effect sizes obtained by the meta-regression approach (Table 4, Fig. 4). The most important difference is, as in subgroup analysis, in the estimation of heterogeneity. Although both meta-regression and the hierarchical model result in the same point estimate for heterogeneity of $\hat{\tau}^2 = 0.02$, the Bayesian approach accounts for uncertainty in this value with a 95% CrI (0.01, 0.07) and provides estimates of the ORs with slightly wider confidence intervals.

One advantage of conducting network meta-analysis as a hierarchical model compared with a meta-regression approach is that ranking of all interventions included in the network is easier. This will be discussed in the next section, on the results from the model that accounts properly for the multi-arm trials.

Table 6 Results of network meta-analysis as hierarchical model for incident diabetes. Log-odds ratios ($\hat{\mu}$) with their standard error $SE(\hat{\mu})$ and odds ratios (OR) with their 95% credible interval (CrI) for all basic and functional contrasts are reported

Comparison	$\hat{\mu}$	$SE(\hat{\mu})$	OR	95% CI for OR
β-blockers versus placebo	0.24	0.10	1.27	(1.04, 1.55)
diuretics versus placebo	0.32	0.10	1.38	(1.13, 1.68)
CCB versus placebo	0.07	0.10	1.07	(0.88, 1.30)
ACE inhibitors versus placebo	−0.09	0.08	0.91	(0.78, 1.07)
ARB versus placebo	−0.19	0.10	0.83	(0.68, 1.01)

Models for Data that Include Multi-arm Trials

When trials involve more than two arms, the network meta-analysis models described in sections "Network Meta-analysis as a Linear Model" and "Network Meta-analysis as a Hierarchical Model" are further complicated for two reasons. The first is the need to account for correlations induced by the fact that multi-arm trials inform more than one comparison. The second is that multi-arm studies are inherently consistent; if A, B, and C are all included within the same study i then, it is plainly the case that $y_{iBC} = y_{iAC} - y_{iAB}$ where y_{ikj} is the effect size in study i for the contrast k versus j. This means that if a study has α_i arms, then only $\alpha_i - 1$ of the $\alpha_i(\alpha_i - 1)/2$ possible comparisons are linearly independent, and so only $\alpha_i - 1$ need to be modelled. This inherent consistency also makes the calculation of the number of degrees of freedom difficult and the formula $N_{\text{comp}} - (T - 1)$ no longerholds (see also section "Statistical Methods to Detect Inconsistency in a Network of Interventions").

Consider the case of three treatments and four studies as presented in Table 7. In study four, only two of the three contrasts need to be included in

Table 7 Data for network meta-analysis with a three-arm trial

Study i	No. arms α_i	Arms/design	Data	Comparison
1	2	A, C	y_{1AC}, s^2_{1AC}	AC
2	2	B, C	y_{2BC}, s^2_{2BC}	BC
3	2	A, B	y_{3AB}, s^2_{3AB}	AB
4	3	A, B, C	y_{4AC}, s^2_{4AC} y_{4BC}, s^2_{5BC} $\text{cov}(y_{4AC}, y_{4AB}) = c$	AC AB

the model as the third effect size y_{iBC} can be simply computed as $y_{iAC} - y_{iAB}$. Thus, the study will contribute *directly* to two out of the three meta-analyses in Fig. 5. The two observed effect sizes y_{iAC} , y_{iAB} are correlated as they both include the common treatment C. This covariance needs to be taken into account in the analysis.

Note that in Table 7, the data for study 4 includes the sample covariance cov(y_{4AC}, y_{4AB}), denoted also as c.The covariance can be estimated from the data as the variance of the outcome in the common arm. For example, if the outcome is continuous and the effect size is the mean difference, it turns out that c is the sample variance of the outcome in the common arm C, that is, $sd_{iC}{}^2/n_{iC}$. When the outcome is dichotomous and the effect size is the ln(OR), the covariance is $c = 1/r_C + 1/(n_c - r_C)$. When the outcome is measured on the risk ratio scale (RR), then the covariance for ln(RR) is $c = 1/r_C - 1/n_C$ and for risk difference it is $c = r_C(n_C - r_C)/n_C{}^3$.

The meta-regression model as presented in section "Network Meta-analysis as a Linear Model" does not account for the dependence between the observations in study 4. Moreover, correlations are present not only in the observations y_{4AC}, y_{4AB} but also in their underlying random effects δ_{4AC}, δ_{4AB}.

Using matrix notation, the meta-regression model in section "Network Meta-analysis as a Linear Model" will have the form

$$\begin{pmatrix} y_{1AC} \\ y_{2BC} \\ y_{3AB} \\ y_{4AC} \\ y_{4AB} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & -1 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mu_{AC} \\ \mu_{AB} \end{pmatrix} + \begin{pmatrix} \delta_{1AC} \\ \delta_{2BC} \\ \delta_{3AB} \\ \delta_{4AC} \\ \delta_{4AB} \end{pmatrix} + \begin{pmatrix} e_{1AC} \\ e_{2BC} \\ e_{3AB} \\ e_{4AC} \\ e_{5AB} \end{pmatrix} \quad (9)$$

To account for the fact that the random errors and the random effects that belong to the same study are correlated, it is assumed that $\boldsymbol{e} \sim N(0, \boldsymbol{S}^2)$ and $\boldsymbol{\delta} \sim N(0, \boldsymbol{T}^2)$ where $\boldsymbol{e}$, $\boldsymbol{\delta}$ are the vectors of the random errors and random effects, $\boldsymbol{S}^2$ is the within-studies variance-covariance matrix (estimated from the data), and $\boldsymbol{T}^2$ is the between-studies variance-covariance matrix (and consists of unknown parameters to be estimated from the model). For the data in Table 7, the within-studies variance-covariance matrix is

$$\boldsymbol{S}^2 = \begin{pmatrix} s^2_{1AC} & 0 & 0 & 0 & 0 \\ 0 & s^2_{2BC} & 0 & 0 & 0 \\ 0 & 0 & s^2_{3AB} & 0 & 0 \\ 0 & 0 & 0 & s^2_{4AC} & c \\ 0 & 0 & 0 & c & s^2_{5AB} \end{pmatrix}$$

whereas the between-studies variance-covariance matrix is

$$\boldsymbol{T}^2 = \begin{pmatrix} \tau^2_{AC} & 0 & 0 & 0 & 0 \\ 0 & \tau^2_{BC} & 0 & 0 & 0 \\ 0 & 0 & \tau^2_{AB} & 0 & 0 \\ 0 & 0 & 0 & \tau^2_{AC} & \mathrm{cov}(\delta_{4AC}, \delta_{4AB}) \\ 0 & 0 & 0 & \mathrm{cov}(\delta_{4AC}, \delta_{4AB}) & \tau^2_{AB} \end{pmatrix}$$

$$= \begin{pmatrix} \tau^2 & 0 & 0 & 0 & 0 \\ 0 & \tau^2 & 0 & 0 & 0 \\ 0 & 0 & \tau^2 & 0 & 0 \\ 0 & 0 & 0 & \tau^2 & \tau^2/2 \\ 0 & 0 & 0 & \tau^2/2 & \tau^2 \end{pmatrix}$$

As discussed before, it is often the case that a common heterogeneity parameter is assumed; that is, $\tau^2_{AC} = \tau^2_{BC} = \tau^2_{AB} = \tau^2$. This assumption offers an advantage in the case of multi-arm studies as it considerably simplifies the between-studies variance-covariance matrix $\boldsymbol{T}^2$. It can be shown that when heterogeneity is equal across comparisons, the covariance of any two random effects is $\tau^2/2$. Then the matrix $\boldsymbol{T}^2$ has τ^2 in the diagonal and $\tau^2/2$ in the cells that refer to pairs of effects from the same study.

Similar considerations need to be made for the hierarchical model. The distributions (7) and (8) apply only to studies $i = 1$, 2 , 3. For $i = 4$ the likelihood of the two-dimensional vector of effect sizes is

$$\begin{pmatrix} y_{4AC} \\ y_{4AB} \end{pmatrix} \times \sim \mathrm{MVN}\left(\begin{pmatrix} \theta_{4AC} \\ \theta_{4AB} \end{pmatrix}, \begin{pmatrix} s^2_4 & \mathrm{cov}(y_{4AC}, y_{4AB}) \\ \mathrm{cov}(y_{4AC}, y_{4AB}) & s^2_5 \end{pmatrix} \right)$$

and the random effects are distributed assuming equal heterogeneities as

$$\begin{pmatrix} \theta_{4AC} \\ \theta_{4AB} \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} \mu_{AC} \\ \mu_{AB} \end{pmatrix}, \begin{pmatrix} \tau^2 & \tau^2/2 \\ \tau^2/2 & \tau^2 \end{pmatrix} \right). \tag{10}$$

The consistency equations remain as presented in section "Network Meta-analysis as a Hierarchical Model".

With arm-specific data and a hierarchical structure, no correlations are needed to account in the likelihood as the observations in arms are independent. For example, if study 4 presents the number of successes r_{4A} , r_{4B} , r_{4C} for a dichotomous outcome out of the total n_{4A} , n_{4B} , n_{4C} randomized, then the likelihood of the data consists of three binomial distributions with event probability parameters p_{4A} , p_{4B} , p_{4C} which parameterized will give two effect sizes θ_4 , θ_5 that correspond to underlying relative effects for treatments A and B compared to C (see section "Meta-analysis as Hierarchical Model"). So, for studies $i = 1$, 2 , 3, the underlying random effects θ_{ijk} follow independent distributions as described in Eq. 8, but the random effects $\theta_{4\text{AC}}$, $\theta_{4\text{AB}}$ from the fourth study will follow the multivariate normal distribution (10).

Technical note: the multivariate normal distribution above can be decomposed into a series of conditional distributions; this offers computational advantages. Distribution (10) can be written as a set of one unconditional and one conditional distribution:

$\theta_{4AB} \sim N(\mu_{AC}, \tau^2)$ and $\theta_{4AC}|\theta_{4AB} \sim N\left(\mu_{AC} + \frac{1}{2}(\theta_{4AB} - \mu_{AB}), \frac{3\tau^2}{4}\right)$

More generally, if a study i has a_i arms that correspond to treatments $T_i = \{A, B, C, D\ldots\}$ in this presented order, the $(a_i - 1)$-dimensional normal distribution of all treatments versus A can be "decomposed" by writing the independent distribution for θ_{iAB}, then the conditional $\theta_{iAC}|\theta_{iAB}$, then $\theta_{iAD}|\theta_{iAB}$, θ_{iAC}, and so on. The distribution of the random effect θ_{iAj} conditional on all "previous" comparisons θ_{iAk} has mean:

$$\mu_{Aj} + \frac{1}{a_i - 1} \sum_{k<j} (\theta_{iAk} - \mu_{Ak})$$

with variance

$$\frac{a_i}{a_i - 1} \frac{\tau^2}{2}$$

where $k < j$ means that comparison Ak was been modelled before Aj.

Application: Network Meta-analysis with Multi-Arm Trials as Hierarchical Model for Incident Diabetes

In this application index i refers to studies ($i = 1, \ldots ,22$). There are 18 studies that compared only two interventions and thus have only two arms ($\alpha_i = 2$), and there are four studies with three arms ($\alpha_i = 3$). The variable $\alpha_i = \{2, 3\}$ needs to be specified for each study i and then a binomial likelihood is assumed for the number of patients in all arms of each study. Using the index j to show the arm (treatment) within a study, the binomial likelihood is written as

$$r_{ij} \sim B(p_{ij}, n_{ij}),\ i = 1, \ldots, 22,\ j \in \{P, BB, D, CCB, ACE, ARB\}$$

The probabilities p_{ij} can be parameterized to model $\alpha_i - 1$ effect sizes as

$\text{logit}(p_{i1}) = u_i$ for the "first" (reference) arm in each study that pertains to treatment j

$\text{logit}(p_{ij}) = u_i + \theta_{ijk}$ for the other arms in the study

The underlying $\ln(OR)$,θ_{ijk}, compares treatments k and j (reported in the first arm) where j , $k \in \{P, BB, D, CCB, ACE, ARB\}$. For the multi-arm trials the correlation between θ_{ijk} and θ_{ijl} , $l \neq j \neq k$in the same trial is taken into account by the conditional mean and variance of their distribution. Table 8 shows the results of fitting this model in WinBUGS against placebo (basic parameters and prior distribution for heterogeneity are as in the application of hierarchical model that does not account for multi-arm trials).

The estimate of common heterogeneity is 0.02 with 95% CrI (0.01, 0.07). Very little change is

Table 8 Results of network meta-analysis as hierarchical model for incident diabetes taking into account multi-arm trials. Log-odds ratios ($\hat{\mu}$) with their standard error $SE(\hat{\mu})$ and odds ratios (OR) with their 95% credible interval (CrI) for all basic and functional contrasts are reported

Comparison	$\hat{\mu}$	$SE(\hat{\mu})$	OR	95% CrI for OR
β-blockers versus placebo	0.24	0.10	1.27	(1.04,1.55)
diuretics versus placebo	0.32	0.10	1.38	(1.13,1.68)
CCB versus placebo	0.07	0.10	1.07	(0.88,1.30)
ACE inhibitors versus placebo	−0.09	0.08	0.91	(0.78,1.07)
ARB versus placebo	−0.19	0.10	0.83	(0.68,1.01)

observed compared with the analyses above in which the correlations between multiple arms were ignored; this is probably due to the fact that multi-arm trials represent only the 18% of our data. All pairwise ORs are presented in Fig. 4.

The posterior deviance from the analysis is $\overline{D} = 53.26$ which, when compared to the number of data points (48), suggests a rather poor fit of the model to the data. The *DIC* of the model was estimated as 91.4.

Network Meta-analysis as a Multivariate Meta-analysis

Multivariate meta-analysis is an extension of meta-analysis that simultaneously synthesizes data on more than one outcome per study. For example, studies which compare antihypertensive interventions might measure the two related outcomes fatal stroke and nonfatal stroke. Some studies will only report fatal or only nonfatal stroke, others will report both. Because these two outcomes are correlated, there are important benefits in analyzing them jointly via multivariate meta-analysis, including improved precision and calculation of confidence regions for both outcomes (Jackson et al. 2011; Riley 2009).

Multiple treatment comparisons reported by multi-arm studies may be viewed in a similar way to multiple outcomes. Specifically, the *basic contrasts* can be considered analogous to different outcomes, where the basic contrasts are the set of necessary comparisons to represent all comparisons under the consistency assumption (e.g., the contrasts Aj of each treatment versus a common reference treatment A). Studies may report on many, allora single basic contrast. In the example of Table 7, the basic contrasts are the contrasts AC and AB. So, study 1 reports on the first "outcome" AC, study 3 reports on the second "outcome" AB, and study 4 reports on both "outcomes."

A departure from the analogy arises for study 2, which compares B and C. This study gives information about a combination of the two "outcomes." To model the BC study, the assumption of consistency is employed. As presented in section "Assumptions Underlying Indirect and Mixed Comparisons" transitivity suggests that the missing arm in a study is missing at random. If study 2 had reported arm A, then the two "outcomes" y_{2AC} , y_{2AB} could have been derived. This suggests a simple imputation strategy, whereby data in the "missing" arm can be created via a data augmentation technique (White 2011).The imputed data are designed to provide minimal information, for example, by giving thema very large variance (for continuous outcomes) or a very small sample size less than one (for dichotomous outcomes). The two effect sizes y_{2AC}, y_{2AB} are correlated, and together they give information about the direct observed BC contrast. The data can be rewritten as in Table 9:

Then, following standard multivariate meta-regression techniques and assuming equal heterogeneities:

$$\begin{pmatrix} y_{1AC} & . \\ y_{2AC} & y_{2BC} \\ . & y_{3AB} \\ y_{4AC} & y_{4BC} \end{pmatrix} = \begin{pmatrix} 1 & . \\ 1 & 1 \\ . & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} \mu_{AC} \\ \mu_{AB} \end{pmatrix} + \begin{pmatrix} \delta_{1AC} & . \\ \delta_{2AC} & \delta_{2BC} \\ . & \delta_{3AB} \\ \delta_{4AC} & \delta_{4BC} \end{pmatrix} + \begin{pmatrix} e_{1AC} & . \\ e_{2AC} & e_{2BC} \\ . & e_{3AB} \\ e_{4AC} & e_{4BC} \end{pmatrix}$$

Table 9 Data for network meta-analysis assuming data augmentation for study 2

Study$i$	No. arms α_i	Arms/ design	Data	Contrast
1	2	A,C	y_{1AC}, s^2_{1AC}	AC
2	2	A imputed, B,C	y_{2AC}, s^2_{2AC} y_{2BC}, s^2_{2BC} cov $(y_{2AC}, y_{2AB}) = c_2$	BC
3	2	A,B	y_{3AB}, s^2_{3AB}	AB
4	3	A,B,C	y_{4AC}, s^2_{4AC} y_{4BC}, s^2_{5BC} cov $(y_{4AC}, y_{4AB}) = c_4$	AC AB

$$\text{with} \begin{pmatrix} \delta_{1AC} & . \\ \delta_{2AC} & \delta_{2BC} \\ . & \delta_{3AB} \\ \delta_{4AC} & \delta_{4BC} \end{pmatrix} \sim NMV\left(\begin{pmatrix} \begin{pmatrix} 0 & . \\ 0 & 0 \\ . & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} \tau^2 & 0 & 0 & 0 \\ 0 & M & 0 & 0 \\ 0 & 0 & \tau^2 & 0 \\ 0 & 0 & 0 & M \end{pmatrix} \end{pmatrix}\right)$$

$$\text{and} \begin{pmatrix} e_{1AC} & . \\ e_{2AC} & e_{2BC} \\ . & e_{3AB} \\ e_{4AC} & e_{4BC} \end{pmatrix} \sim NMV\left(\begin{pmatrix} \begin{pmatrix} 0 & . \\ 0 & 0 \\ . & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} s^2_{1AC} & 0 & 0 & 0 \\ 0 & L_1 & 0 & 0 \\ 0 & 0 & s^2_{3AB} & 0 \\ 0 & 0 & 0 & L_2 \end{pmatrix} \end{pmatrix}\right)$$

$$L_1 = \begin{pmatrix} s^2_{2AC} & c_2 \\ c_2 & s^2_{2BC} \end{pmatrix}, L_2 = \begin{pmatrix} s^2_{4AC} & c_4 \\ c_4 & s^2_{5BC} \end{pmatrix} \text{ and } M = \begin{pmatrix} \tau^2 & \tau^2/2 \\ \tau^2/2 & \tau^2 \end{pmatrix}.$$

It has been shown that the choice of the basic contrasts and the data augmentation parameters do not impact on the estimation of the effects. The model can be fitted using the **mvmeta** command in STATA, providing estimates of the means and standard errors for the basic parameters μ_{AC} and μ_{BC}. As described earlier, combinations of the basic parameters can give estimates of all functional parameters through application of the consistency equations and uncertainty in these can be obtained by incorporating covariances between the estimates.

Application: Network Meta-analysis for Incident Diabetes as a Multivariate Meta-analysis

The use of standard multivariate meta-regression requires that all studies have data for the treatment that has been chosen as reference. When none of the treatments of the network is common to all studies (as in the current example), one of the treatments can be chosen to be the common reference treatment and then the data augmentation technique is applied. More specifically, choosing placebo as reference implies that in studies without a placebo arm, we need to "impute" data for a very small sample size for an assumed placebo arm, and here the values $r_{iP} = 0.001$ and $n_{iP} = 0.01$ are used. Then all studies will report on the relative effectiveness of the included treatments versus placebo, y_{iPj} where $j = \{BB, D, CCB, ACE, ARB\}$. The sample variance-covariance matrix S of all y_{iPj} needs to be specified. As the outcome is measured using the (OR), the variances of all observations are calculated using the formula:

$$s^2_{iPj} = \frac{1}{r_P} + \frac{1}{n_P - r_P} + \frac{1}{r_j} + \frac{1}{n_j - r_j}$$

and the covariances are calculated as

$$\text{cov}\left(y_{iPj}, y_{iPk}\right) = \frac{1}{r_P} + \frac{1}{n_P - r_P}$$

The variance-covariance matrix of the random effects can be modelled in various ways. The most flexible structure is to estimate different heterogeneity variances τ^2_{Pj} for each comparison (j vs. P). In the analyses that follow, a much more restricted structure is used, following the assumption of a common heterogeneity variance as has been used in previous analyses in the chapter. This sets $\tau^2_{Pj} = \tau^2$ so all covariances between random effects are $\tau^2/2$. This model can be implemented in STATA using the **mvmeta** command with the

Table 10 Results of network meta-analysis as multivariate meta-analysis for incident diabetes. Log-odds ratios ($\hat{\mu}$) with their standard error $SE(\hat{\mu})$ and odds ratios (OR) with their 95% confidence interval (CI) for all basic contrasts are reported

Comparison	$\hat{\mu}$	$SE(\hat{\mu})$	OR	95% CI for OR
β-blockers versus placebo	0.21	0.08	1.24	(1.05,1.44)
diuretics versus placebo	0.28	0.08	1.32	(1.12,1.56)
CCB versus placebo	0.04	0.08	1.04	(0.89,1.21)
ACE inhibitors versus placebo	−0.12	0.07	0.88	(0.77,1.10)
ARB versus placebo	−0.19	0.09	0.83	(0.70,0.98)

option `bscov()`, which gives the results of Table 10.

Estimates for all functional comparisons can be derived with the use of consistency equations. There are small differences between the results of this approach with the corresponding results of the hierarchical model. Using the restricted maximum likelihood estimator in mvmeta results in $\hat{\tau}^2 = 0.01$, which is the same as the heterogeneity estimated in the hierarchical model.

Assumptions of Network Meta-analysis

As presented in section "Estimating Inconsistency in Mixed Comparisons," inconsistency in a network can manifest as a disagreement between different sources of evidence for the same comparison and can be identified statistically. For example, an indirect estimate of *A* versus *B* via a treatment *C* can be in conflict with the direct estimate or with another indirect estimate, e.g., *A* versus *B* via a treatment *C*.

Both the likelihood of transitivity (based on clinical and epidemiological considerations) and any evidence of (in)consistency (based on statistical considerations) should be evaluated in a network as part of a network meta-analysis. Conceptual evaluation involves a priori judgements about the comparability of the studies across comparisons with respect to the distribution of potential confounders, considering whether treatments were all given for the same indication and considering whether anchor treatments are equivalent. Such judgements should be made ideally before the outcome data are extracted but after the studies and their characteristics are collected.

Although transitivity and consistency are interwoven concepts and are often thought of as one, it can be useful to consider them separately for ease of evaluation. Consider, for instance, the network presented in Fig. 6 where all treatments have been compared with placebo but not with each other. In this network, each given comparison is informed by a single source of evidence (there are no closed loops in the network), and therefore it is impossible to observe inconsistency by statistical means. However, network meta-analysis can be performed and will yield estimates for the relative effects of all active treatments (which in this case are only indirect estimates). These estimates are valid only if transitivity can be assumed for the common "anchor" treatment (placebo). Therefore, judgements about the plausibility of transitivity should be done for the entire network as indirect evidence is derived for all treatments irrespectively of whether they belong to "closed loops" or not.

Statistical Methods to Detect Inconsistency in a Network of Interventions

The assumption of consistency can be evaluated statistically in a full network by extending the idea outlined in section "Estimating inconsistency in Mixed comparisons". A network often comprises several closed loops (triangles, quadrilaterals, etc.) which bring together evidence for the same comparison from direct and various indirect routes. Within each one of these closed loops, the consistency assumption, seen as agreement between direct and indirect estimates, can be evaluated. A first approach is therefore to evaluate

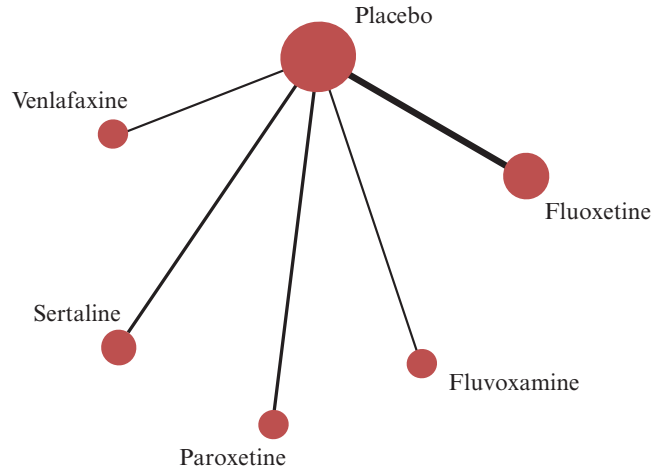


Fig. 6 Plot of a network for efficacy of pharmacotherapeutic agents for anxiety disorders in children and adolescents. The size of the nodes is proportional to the number of studies that evaluate each intervention and the thickness of the lines is proportional to the frequency of each comparison in the network (Uhtman and Abdulmalik 2010)

inconsistency in all loops by calculating the loop-specific inconsistency factors $\hat{IF}$, their confidence intervals, and a z-test for each one. The $\hat{IF}s$ of a network can be presented in a forest plot like graph where deviations from the consistency assumption would be reflected in loops with confidence intervals incompatible with zero. Note that not all loops need to be presented and tested; for example, if a quadrilateral consists of two triangles inconsistency needs to be evaluated only in the triangles. Consider, for example, the network in Fig. 7. Inconsistency can be evaluated in *ABC* and *ADB* loops; if these are consistent then the quadrilateral *ADBC* would be consistent as well.

The loop-based approach is simple and can be useful for identifying loops that deviate from consistency, but has important limitations. An obvious problem is that the loop-specific tests are not independent as they share groups of studies. Consider, for example, the network in Fig. 7 and imagine that the *AB* comparison is informed by a single study in which an unobserved characteristic produced an estimate very different from what would be expected in the other studies. Then both *ABC* and *ADB* loops will present inconsistency because the respective $\hat{IF}s$ share the same deviant *AB* study.

The loop-based approach does not provide a network-specific estimate of the inconsistency. The multiple dependent tests cannot be summarized into a global network-specific test. It is also unclear how to treat multi-arm trials, which are

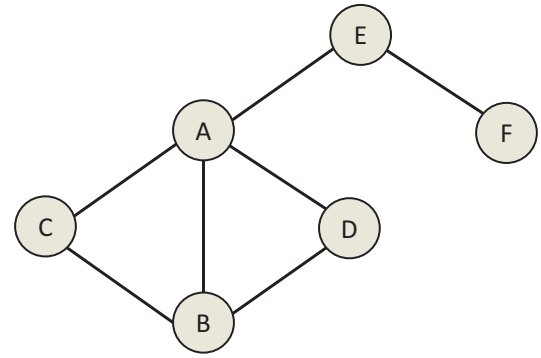


Fig. 7 Fictional network of interventions with two triangular loops

inherently consistent. Because of the dependence between the loops and the multiple testing nature of the approach, the results should be interpreted with caution; the absence of inconsistent loops may be reassuring for the assumption of consistency (notwithstanding the lack of power of such tests), but the presence of statistically significant loops cannot be used to infer the magnitude of inconsistency in a network.

In the special case where the loops share a single comparison as in Fig. 7, a chi-squared test can be applied (Caldwell et al. 2010). For the same comparison *AB*, there are three estimates; the direct estimate $\hat{\mu}^{D}_{AB}$ the two indirect estimates via *C* and *D*, $\hat{\mu}^{I}_{ABviaC}$ and $\hat{\mu}^{I}_{ABviaD}$, respectively, with their estimated variances noted as $\hat{v}^{D}_{AB}, \hat{v}^{I}_{ABviaC}, \hat{v}^{I}_{ABviaD}$. The mixed estimate $\hat{\mu}^{M}_{AB}$ is

the weighted average of the three estimates with weights being the inverse of the variance. To test both *ABC* and *ADB* loops (and therefore provide a global test for the network) the following chi-squared test can be applied

$$\frac{\left(\hat{\mu}^{D}_{AB}-\hat{\mu}^{M}_{AB}\right)^2}{\hat{v}^{D}_{AB}}+\frac{\left(\hat{\mu}^{I}_{ABviaC}-\hat{\mu}^{M}_{AB}\right)^2}{\hat{v}^{I}_{ABviaC}}+\frac{\left(\hat{\mu}^{I}_{ABviaD}-\hat{\mu}^{M}_{AB}\right)^2}{\hat{v}^{I}_{ABviaD}}\sim \mathrm{X}^2_2$$

This can be generalized to combine testing for disagreement between direct estimate and l independent indirect sources; the weighted sum of the difference of each estimate from the mixed estimate will follow a chi-squared distribution with l degrees of freedom.

The results of the loop-based approach can vary substantially depending on the method used to derive the pairwise estimates and their variances. In the presence of heterogeneity, the uncertainty of $\hat{IF}$ will be larger in a random effects analysis compared with a fixed-effect analysis, and therefore there will be less chance of identifying statistically significant inconsistencies. The random effects approach will also give different results depending on which method will be used to estimate the heterogeneity parameter τ (e.g., method of moments, restricted maximum likelihood). Some approaches will give larger estimates than others, resulting in different estimates for the uncertainty of $\hat{IF}$. Moreover, the estimated pairwise variances will change depending on whether the same or different heterogeneity parameters are assumed in the loop.

There is currently limited empirical evidence about the occurrence of statistical inconsistency. A study evaluated 112 triangular networks of which only 16 were found inconsistent (Song et al. 2011). O'Regan et al. (2009) empirically evaluated the agreement between indirect and mixed estimates that appear in networks of at least four treatments. Using a fixed-effect approach, they concluded that the two indirect and mixed estimates did not show important differences, although the 51 comparisons they examined came from only seven reviews.

Approaches that evaluate inconsistency globally in a network rather than testing each loop have gained in popularity but are typically cumbersome to apply and have limitations. For network models fitted within a Bayesian framework, the consistency assumption can be evaluated by comparing a model that assumes consistency with one that does not, using the *DIC* (Spiegelhalter et al. 2002). The model without consistency is the model described in section "Consistency Models" but without the consistency equations to derive indirect and mixed estimates. The inconsistency model relies only on direct evidence and is equivalent to a series of pairwise meta-analyses (usually assuming, however, that they share the same heterogeneity parameter). The assumption of consistency is challenged when the inconsistency model presents, for the same data, a better trade-off between model fit and complexity; this is the case when the DIC for the inconsistency model is lower to the DIC for the consistency model by more than three units. An important drawback with this method is that results may depend on the parameterization of the multi-arm trials, from which only some of the study-specific effect sizes enter the model. Approaches that simultaneously test and account for inconsistency are discussed in the section "Inconsistency models".

Application: Statistical Evaluation of Inconsistency in Each Loop of Incident Diabetes Network

The network includes 16 "triangles" that can be evaluated for inconsistency. For the calculation of all inconsistency factors, the formulae of section 3.3.3 is employed. Then the estimates with their 95% CI can be plot in a forest plot. The pairwise effect sizes were estimated using the random effects model assuming different and loop-common heterogeneity parameters. There are no important differences between the two forest plots; both include two inconsistent loops.

The hierarchical model is fitted as described in section "Network Meta-analysis as a Hierarchical Model" but omitting the consistency equations (i.e., an inconsistency model); i.e., this is

essentially a sequence of pairwise meta-analyses. The value of the posterior deviance was $\overline{D} = 50.85$ and DIC = 93.6. Comparing the $\overline{D}$ value to that obtained from the consistency model, since the difference in DIC is smaller than three points, this suggests that the inconsistency model fits the data better and might also be the most parsimonious model.

Inconsistency Models

Two major approaches have been proposed so far to address inconsistency. The first approach was proposed in (Lu and Ades 2006) and is based on the idea that inconsistency is a property of closed loops and a network can have as many inconsistencies as functional parameters. Recently, an approach has been proposed which extends the idea of inconsistency: it does not apply only to the disagreement between direct and indirect estimates in a loop but also disagreement between studies that report the same comparison but include different sets of treatments. The two approaches are outlined below, starting from the data in Table 7.

The loop-based inconsistency model assumes that inconsistency arises when the consistency equations between functional and basic parameters do not hold. Hence, an obvious solution is to "relax" the assumption by adding an extra term to account for inconsistencies. In the example of Table 7, there are two basic parameters μ_{AC} , μ_{AB} and one functional $\mu_{BC} = \mu_{AC} - \mu_{AB}$. This reflects the closed loops ABC. Inconsistency in this loop can be accounted for if it is assumed that

$$\mu_{BC} = \mu_{AC} - \mu_{AB} + w_{ABC}$$

where w_{ABC} measures the amount of inconsistency in the loop. The term is also called an inconsistency factors and in fact in the absence of multiple correlated loops is analogous to the simple $\hat{IF}$. In complex networks where many inconsistency factors exist, the parameters w_{jkf} are assumed to be randomly distributed with expectation zero:

$$w_{jkf} \sim N\left(0, \sigma^2\right)$$

The variance σ^2 is often referred to as the *inconsistency variance* in analogy with the heterogeneity variance τ^2 in the distribution of the study-specific random effects $\delta_i \sim N(0, \tau^2)$. The inconsistency σ^2 describes the amount of variability across loops in the conflict between direct and indirect evidence. Monitoring the individual w_{jkf} s for large values will reveal loops with important inconsistency, whereas comparison of σ^2 to τ^2 will show how much inconsistency exists compared with the heterogeneity.

As the degrees of freedom in a network describe the number of functional parameters, there are $N_{\text{com}} - (T - 1)$ many inconsistency factors. Problems arise with this approach when there are multi-arm trials. The ABC trials in Table 7 are inherently consistent, and therefore the BC comparison reported in these studies does not contribute to the inconsistency as much as the BC comparison in an independent study. Lu and Ades suggested adjusting the inconsistency degrees of freedom to $ICDF = N_{\text{com}} - (T - 1) - S$ where S is the number of independent inconsistency relations in which the corresponding parameters are supported by no more than two independent sources of evidence. In practice, S is the number of functional comparisons where two out of the three parameters are only estimated in multi-arm trials.

The difficulties in fully defining loop inconsistency when there are multi-arm studies motivated the concept of "design inconsistency." Design inconsistency reflects the belief that studies which include different treatments might give different estimates for the same comparison. For example, an AB study and ABC study might provide different estimates because of their different design. Design inconsistency can be thought of as a special case of source-specific heterogeneity: variation between the estimates for the same comparison due to differences in the total treatments included. In the data of Table 7, this means adding an inconsistency factor for the disagreement between the three estimates in the ABC study and the AB , AC, and BC studies. The model

with both loop and design inconsistency has $N_{\text{Comp}\times\text{Design}} - (T-1)$ inconsistency factors, where $N_{\text{Comp}\times\text{Design}}$ is the number of independent comparisons per design. In Table 7 there is one independent comparison for each two-arm trial and two independent comparisons for the three-arm trials. This results in a total of three inconsistency factors for the network. These inconsistency factors are comparison-specific and are attached to every study reporting that comparison. For instance, one inconsistency factor is attached to each AB , AC, and BC study, respectively, (w_{AB} , w_{AC} , w_{BC}). As the inconsistency factors derived in this way are independent, they can be summarized in a single test for the entire network (see White 2011 for details).

One further approach for detecting inconsistency in a network meta-analysis is "node splitting" (Dias et al. 2010) where a "node" refers to each summary effect generated from the network meta-analysis. This approach is based on the separation of the information contributing to each node into the direct and indirect evidence, within a single model. The node-splitting approach allows the analyst to split the network-wide information contributing to the summary estimate into the evidence directly comparing B versus $C(\hat{\mu}^{D}_{BC})$ and all the remaining "indirect" evidence for B versus C ($\hat{\mu}^{I}_{BC}$) after the studies directly comparing B to C have been removed. The extent of agreement between the direct and indirect estimates defines the magnitude of consistency. Note that this is a computationally intensive approach involving models that can be difficult to parameterize; care should be taken to ensure that multi-arm trials are handled correctly and to ensure that split nodes are actually from contrasts contributing to suspect loops.

Application: Hierarchical Inconsistency Model for Network Meta-analysis in Incident Diabetes

The application of a hierarchical inconsistency model requires careful choice of the basic parameters μ_{Aj} and the inconsistency factors w_{jkf}, as well as the appropriate parameterization of multi-arm trials. First, all basic contrasts need to be informed directly from at least one study. Choosing placebo as reference treatment (A) satisfies this condition, because all other treatments are compared directly with placebo in at least one study. Second, the four multi-arm trials included in the data may modify the number of ICDF that should be included in the model. However, as all consistency equations are informed by at least three independent sources of evidence, it is

$$\begin{aligned} ICDF &= N_{\text{com}} - (T-1) - S \\ &= 14 - (6-1) - 0 = 9 \end{aligned}$$

The consistency relations can be relaxed to include the nine inconsistency parameters:

$$\mu_{D-BB} = \mu_{D-P} - \mu_{BB-P} + w_{P-D-BB}$$

$$\mu_{CCB-D} = \mu_{CCB-P} - \mu_{D-P} + w_{P-CCB-D}$$

$$\mu_{CCB-BB} = \mu_{CCB-P} - \mu_{BB-P} + w_{P-CCB-BB}$$

$$\mu_{ACE-CCB} = \mu_{ACE-P} - \mu_{CCB-P} + w_{P-ACE-CCB}$$

$$\mu_{ACE-D} = \mu_{ACE-P} - \mu_{D-P} + w_{P-ACE-D}$$

$$\mu_{ACE-BB} = \mu_{ACE-P} - \mu_{BB-P} + w_{P-ACE-BB}$$

$$\mu_{ARB-CCB} = \mu_{ARB-P} - \mu_{CCB-P} + w_{P-ARB-CCB}$$

$$\mu_{ARB-D} = \mu_{ARB-P} - \mu_{D-P} + w_{P-ARB-D}$$

$$\mu_{ARB-BB} = \mu_{ARB-P} - \mu_{BB-P} + w_{P-ARB-BB}$$

where $w_{Pjk} \sim N(0, \sigma^2)$ for j , $k = \{BB, D, CCB, ACE, ARB\}$. The rest of the model is the same with the consistency hierarchical model (accounting for multi-arm trials). Moreover, contrasts that are informed only from multi-arm trials need to be expressed in model parameters. Such contrasts are β-blockers versus placebo, included in a D-BB-P study, and ACE inhibitors versus CCB, included in two ACE-CCB-BB, and one ACE-CCB-D studies. Since the model considers treatment 1 as baseline treatment (A) of each study, we need in the data for the D-BB-P trial placebo to be the first treatment and for the other three studies CCB or ACE inhibitors.

Table 11 Results of inconsistency hierarchical model for network meta-analysis for incident diabetes. Inconsistency factors (w), log-odds ratios ($\hat{\mu}$) with their standard error $SE(\hat{\mu})$, and odds ratios (OR) with their 95% credible interval (CrI) for all basic and functional contrasts are reported. Missing values of w correspond to basic contrasts or functional contrasts without direct estimates available

Comparison	w_{Pjk}	$\hat{\mu}$	$SE(\hat{\mu})$	OR	95% CI for OR
β-blockers versus placebo	–	0.23	0.11	1.26	(1.03, 1.62)
diuretics versus placebo	–	0.31	0.10	1.36	(1.13, 1.71)
CCB versus placebo	–	0.06	0.10	1.06	(0.89, 1.32)
ACE inhibitors versus placebo	–	−0.13	0.08	0.88	(0.75, 1.03)
ARB versus placebo	–	−0.20	0.10	0.82	(0.66, 1.00)
diuretics versus β-blockers	−0.02	0.08	0.12	1.08	(0.86, 1.36)
CCB versus diuretics	0.00	−0.25	0.11	0.78	(0.62, 0.97)
CCB versus β-blockers	−0.01	−0.17	0.11	0.84	(0.68, 1.04)
ACE inhibitors versus CCB	0.01	−0.19	0.11	0.83	(0.65, 1.00)
ACE inhibitors versus diuretics	0.01	−0.44	0.11	0.65	(0.50, 0.78)
ACE inhibitors versus β-blockers	0.04	−0.36	0.12	0.70	(0.53, 0.85)
ARB versus ACE inhibitors	–	−0.07	0.12	0.93	(0.73, 1.18)
ARB versus CCB	0.00	−0.26	0.12	0.77	(0.59, 0.95)
ARB versus diuretics	−0.01	−0.51	0.13	0.60	(0.45, 0.76)
ARB versus β-blockers	0.02	−0.43	0.13	0.65	(0.49, 0.81)

Table 11 shows the results of fitting this model in WinBUGS employing a half-normal prior distribution on the inconsistency variance σ^2 (the same as for the heterogeneity τ^2).

Heterogeneity and inconsistency variances were estimated as 0.02 and 0.01, respectively, with 95% CrI (0, 0.06) and (0, 0.13), respectively. Some w-factors are quite large in relation to the treatment effect estimates, indicating that there is probably some inconsistency in the network. Note this is in agreement with the loop-specific approach. The loop placebo versus ACE inhibitors versus β-blockers presents the largest inconsistency value (0.04) followed by the loops placebo versus ARB versus β-blockers (0.02) and placebo versus diuretics versus β-blockers (−0.02). The first two loops were also identified as inconsistent in Fig. 8, and the last was appeared marginally consistent. There are no large differences in the point estimates of the summary ORs compared to those from the consistency model. However, the 95% CrI from the inconsistency model are wider to account for inconsistency.

The DIC of the model was 92.1 and $\overline{D} = 53.15$ showing that accounting for inconsistency does not improve the fit of the model as the consistency model resulted in almost the same values.

Exploring Heterogeneity and Inconsistency: Network Meta-regression

When heterogeneity is found in a pairwise meta-analysis, subgroup analysis or meta-regression are employed to explore possible sources. Network meta-regression is an extension of network meta-analysis to include covariates and can be used to explore heterogeneity and/or inconsistency. Covariates typically include study-specific variables such as setting or length of follow-up, within-trial bias characteristics such as the quality of randomization, of allocation concealment and blinding, or patient-level characteristics such as age or sex. Meta-regression is equivalent to subgroup analysis for dichotomous or categorical explanatory variables. Characteristics such as differences in baseline risk (if there is a common comparator) and sample size (as a single proxy for study quality) can also be considered.

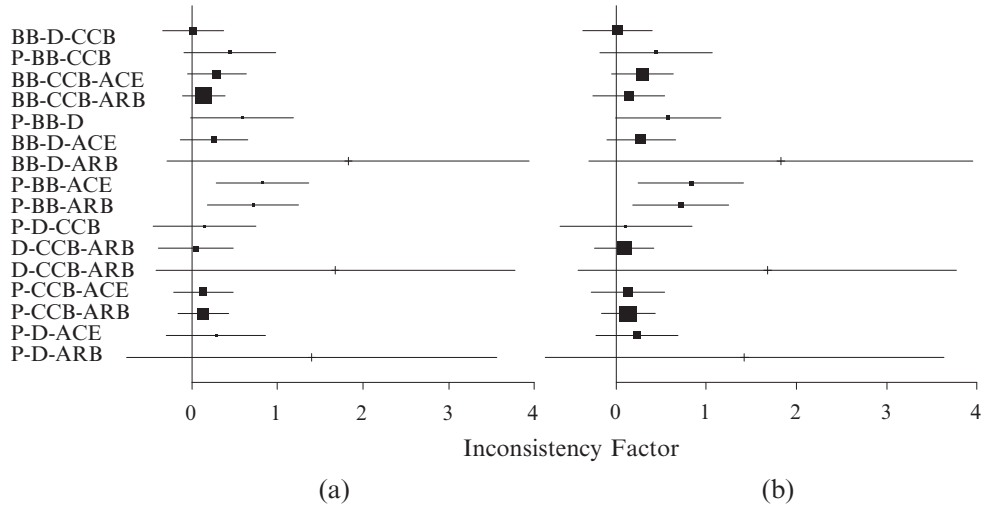


Fig. 8 Inconsistency factors of all triangles of incident diabetes network with (**a**) a different heterogeneity estimate for each comparison and (**b**) with a common heterogeneity estimate within each triangle. Triangles with statistically significant inconsistency factors (their 95% CI does not include 0) are considered as inconsistent

The network meta-regression model as a hierarchical model is

$$y_{ijk} \sim N\left(\theta^*_{ijk}, s^2_{ijk}\right)$$

$$\theta^*_{ijk} = \theta_{ijk} + b_{ijk}C_{ijk}$$

$$\theta_{ijk} \sim N\left(\mu_{jk}, \tau^2_r\right)$$

where b_{ijk} are the regression coefficients for study i and comparison jk and C_{ijk} the explanatory variable. The regression coefficients can be assumed to be fixed across studies ($b_{ijk} = \beta_{jk}$) or, if there are many studies per comparison, as exchangeable across studies ($b_{ijk} \sim \mathrm{N}(\beta_{jk}, \gamma^2)$). The model can be applied to multi-arm trials and also extended to account for inconsistency as described in previous sections.

Consistency can be imposed for the regression coefficients by choosing a reference treatment A and defining $\beta_{jk} = \beta_{Ak} - \beta_{Aj}$ (Cooper et al. 2009). To improve power, the independent β_{Aj} can be assumed exchangeable; $\beta_{Aj} \sim \mathrm{N}(\mathrm{B}, \varphi^2)$. Adjusting for factors that can vary across comparisons may reduce heterogeneity and improve the likelihood of transitivity. The importance and impact of the adjustment can be judged by monitoring changes in the heterogeneity variance (compare τ^2_r to τ^2) and inconsistency variance (compare σ^2_r to σ^2), by monitoring the magnitude and significance of the coefficients β_{jk} and by comparing the goodness of fit and parsimony of adjusted and unadjusted models using DIC and $\overline{D}$.

Network meta-regression suffers from the same problems with simple meta-regression. These include ecological bias when aggregated patient-level data are used as covariates, low power with few studies and high false-positive rates if heterogeneity not explained by the covariances is ignored (Higgins and Thompson 2004).

Adjusting for bias in a network of interventions offers the advantage of increased power compared with traditional meta-analysis sensitivity analysis, because the regression coefficients share information via the consistency equations. Suppose, for example, that comparison B versus C is informed by very few studies, or by studies that all have the same characteristic (e.g., they all have poor allocation concealment). Then, conducting sensitivity analysis or adjusting the meta-analysis result of BC for allocation concealment is suboptimal or

impossible. However, when these studies are part of a network meta-regression model, the bias coefficient β_{BC} for allocation concealment is linked to the other regression coefficients via $\beta_{BC} = \beta_{AC} - \beta_{AB}$ and $\beta_{Aj} \sim \mathrm{N}(\mathrm{B}, \varphi^2)$.

A special application of network meta-regression is to address small study effects in a network of interventions. The association between sample size, effect size heterogeneity, and the probability of publication (which is often manifested as funnel plot asymmetry) has long been a challenging issue in meta-analysis. In a pairwise meta-analysis, the presence of small study effects (possibly due to publication bias) has been explored by regressing the underlying effect on a measure of the study precision. The same approach applies to networks of interventions to explore situations where comparisons that do not give significant results may be underrepresented or missing in the network and their relative effectiveness will be informed primarily by the indirect evidence. The covariate C_{ijk} can be the sample standard error, variance, or inverse of sample size (references). However, significant associations between effect sizes and precision can be taken only as an indication of publication bias, as other explanations, including genuine heterogeneity, are possible. As publication bias and selective reporting will affect interventions and comparisons in different ways depending on the clinical context, the problem of selection bias in the network should be considered carefully. Further methodological development is needed to better address selection bias in network meta-analysis.

Because network meta-analysis combines studies that compare a treatment against a variety of comparators, it enables researchers to explore biases that are not identifiable in a head-to-head meta-analysis. "Optimism bias" associated with the use of novel interventions has been a concern difficult to address in a pairwise meta-analysis (Djulbegovic et al. 2011; Heres et al. 2006; Soares et al. 2005). However, in a network of interventions, the same treatment C can be the newer and hence the "favored" in a comparison A versus C but the older in another comparison B versus C. This enables us to explore apparent changes in the effectiveness of C because of optimism (Salanti et al. 2010).

Application: Network Meta-regression for Incident Diabetes Using Year of Publication as Covariate

An network meta-regression analysis of the incident diabetes data set will investigate whether differences in the publication year of included studies have an impact on the estimated treatment effects, and hence whether they can explain any of the heterogeneity and inconsistency of this network. Two meta-regression models will be used; one estimating a common fixed coefficient across all studies and all treatment comparisons and a second imposing consistency in coefficients. More specifically, in the first model, it is assumed that $b_{ijk} = B(i = 1, \ldots, 22)$, and a vague normal prior distribution on the fixed coefficient $\sim N(0, 10000)$is employed. In the second model, the coefficients are assumed to be consistent $b_{ijk} = \beta_{Ak} - \beta_{Aj}$ $(A = P)$ and exchangeable $\beta_{Aj} \sim \mathrm{N}(\mathrm{B}, \varphi^2)$ $(j = \{BB, D, CCB, ACE, ARB\})$, where B and φ^2 are the mean and variance, respectively, of the distribution of all β_{Aj} with normal $(B \sim N(0, 10000))$ and half-normal prior distributions. In both models a covariate $\left(C_i - \overline{C}_i\right)$ is used instead of C_i (the year of publication of study i) for computational reasons (e.g., convergence of the models), where $\overline{C}_i$ is the mean publication year.

The estimate of the fixed regression coefficient from the first model (B) was -0.01, corresponding to an odds ratio that is $e^{-0.01} = 0.99$ times smaller for each 1 year later of publication. However, the 95% CrI of B is $(-0.03, 0.01)$ implying that there is no statistically significant effect of study publication year on treatments' effectiveness.

The same inference is derived from the second meta-regression model, which estimates the mean (B) of distribution of regression coefficients' to be -0.02 $(-0.07, 0.01)$ with variance $\varphi^2 < 0.001$.

Table 12 Medians and 95% CrI of regression coefficients for comparisons of all treatments versus placebo estimated by network meta-regression model for incident diabetes with consistent and exchangeable coefficients

Comparison	β	95% CrI
β-blockers versus placebo	−0.02	(−0.06, 0.01)
diuretics versus placebo	−0.02	(−0.06, 0.01)
CCB versus placebo	−0.03	(−0.07, 0.02)
ACE inhibitors versus placebo	−0.03	(−0.08, 0.01)
ARB versus placebo	−0.03	(−0.10, 0.03)

Table 13 Results of network meta-regression model with a common fixed coefficient for incident diabetes using year of publication as covariate. Log-odds ratios ($\hat{\mu}$) with their standard error $SE(\hat{\mu})$ and odds ratios (OR) with their 95% credible interval (CrI) for all comparisons are reported

Comparison	$\hat{\mu}$	$SE(\hat{\mu})$	OR	95% CrI for OR
β-blockers versus placebo	0.23	0.09	1.26	(1.06,1.50)
diuretics versus placebo	0.26	0.09	1.29	(1.07,1.56)
CCB versus placebo	0.06	0.09	1.07	(0.90,1.26)
ACE inhibitors versus placebo	−0.10	0.07	0.91	(0.78,1.05)
ARB versus placebo	−0.16	0.10	0.86	(0.70,1.04)

Table 14 Results of network meta-regression model with consistent and exchangeable coefficients for incident diabetes using year of publication as covariate. Log-odds ratios ($\hat{\mu}$) with their standard error $SE(\hat{\mu})$ and odds ratios (OR) with their 95% credible interval (CrI) for all comparisons are reported

Comparison	$\hat{\mu}$	$SE(\hat{\mu})$	OR	95% CrI for OR
β-blockers versus placebo	0.25	0.10	1.28	(1.06, 1.55)
diuretics versus placebo	0.30	0.10	1.34	(1.11, 1.64)
CCB versus placebo	0.05	0.10	1.05	(0.86, 1.29)
ACE inhibitors versus placebo	0.09	0.10	1.09	(0.91, 1.33)
ARB versus placebo	−0.06	0.09	0.94	(0.79, 1.12)

Table 12 shows the consistent coefficients (β_{Aj}) of all treatments versus placebo.

The estimated treatment effects by the two models are presented in Tables 13 and 14.

Both meta-regression models resulted in heterogeneity estimates $\hat{\tau}^2 = 0.02$, the same as for the consistency hierarchical model (accounting for multi-arm trials), showing that year of publication as a covariate does not explain adequately the heterogeneity in the network.

The meta-regression with a fixed coefficient also does not improve the fit of the model ($\overline{D} = 53.85$, DIC = 92.2) compared with the hierarchical consistency model without any covariates, while the model with consistent coefficients shows a slightly better fit ($\overline{D} = 51.57$, DIC = 91.3).

The inconsistency model (as described in section "Consistency Models" but omitting the consistency equations) was also fitted with a fixed coefficient to investigate if differences in year of publication can explain the identified inconsistency. The value of the posterior deviance was $\overline{D} = 50.43$ and DIC = 93.4, same with the inconsistency model not including any covariates. However, using the estimates of this model, the two inconsistent loops ACE-BB-P and ARB-BB-P become consistent with IF = 0.20 (−0.44, 0.85) and 0.18 (−0.53,

0.89) implying that year of publication is a possible explanation of inconsistency.

Numerical and Graphical Presentation of Results from Network Meta-analysis

Network meta-analysis involves many treatments and consequently results in a plethora of pairwise effect sizes. When presenting results from a network meta-analysis, it is useful to show both the direct and the mixed estimates along with their 95% confidence intervals and comment on any disagreements between them (e.g., as in Fig. 3). In a consistency model, all pairwise comparisons are possible and the effect sizes are often presented in the form of a "league table" or in a forest plot against a common comparator (see, e.g., Fig. 4). Presentation of the results using predictive

Table 15 Ranking probabilities for all treatments of incident diabetes. Results are based on the consistency hierarchical model (accounting for multi-arm trials)

Order	Placebo	β-Blockers	Diuretics	CCB	ACE inhibitors	ARB
1	0.01	0.00	0.00	0.00	0.22	0.77
2	0.07	0.00	0.00	0.02	0.71	0.20
3	0.65	0.00	0.00	0.27	0.06	0.02
4	0.27	0.01	0.01	0.70	0.01	0.00
5	0.01	0.79	0.20	0.01	0.00	0.00
6	0.00	0.20	0.80	0.00	0.00	0.00

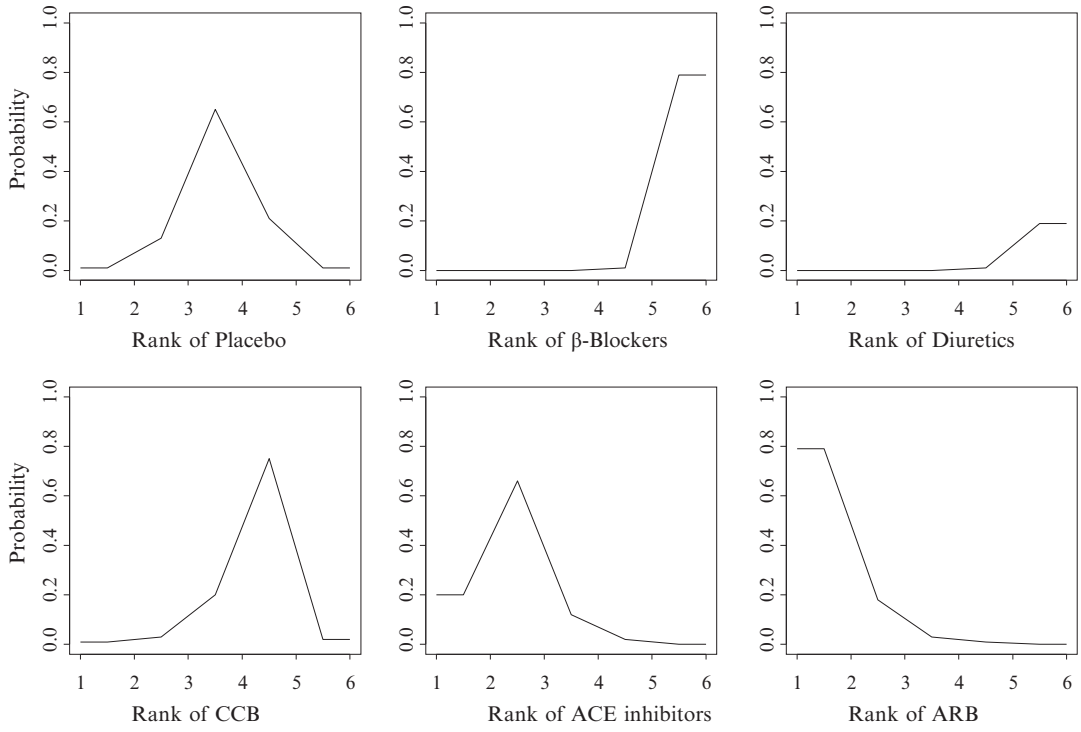


Fig. 9 "Rankograms" for all treatments of incident diabetes. Results are based on the consistency hierarchical model (accounting for multi-arm trials)

intervals, though infrequent, best conveys the uncertainty due to heterogeneity.

Ranking measures and probabilities have become popular as they provide an understandable gateway to the results, particularly when there are many competing treatments. The probability of each treatment being the best is often calculated when the network model is fitted within the Bayesian framework. Methods are also available for similar ranking of treatments in a frequentist framework (White 2011). The probability of being the best treatment has the disadvantage that it does not reflect spread of rankings for the treatments and may thus be misleading. An obvious solution is to calculate the probabilities for all ranks. The probability of each treatment to achieve each possible rank can be plotted to yield "rankograms." Presentation of the cumulative ranking curves in a single plot and a numerical summary of the area below the cumulative raking curve for each treatment is useful as it gives a clear ordering of all treatments based on a summary of the rank probabilities. A review of graphical and numerical methods along with software code are presented in (Chaimani et al. 2013).

Table 16 Numerical summary of area below the cumulative raking curve for all treatments of incident diabetes. Results are based on the consistency hierarchical model (accounting for multi-arm trials)

Placebo	0.59
β-Blockers	0.16
Diuretics	0.04
CCB	0.46
ACE inhibitors	0.81
ARB	0.96

Application: Presentation of Results for Incident Diabetes

The results of the consistency hierarchical model (accounting for multi-arm trials) will be used to illustrate the use of rankograms. The hierarchical

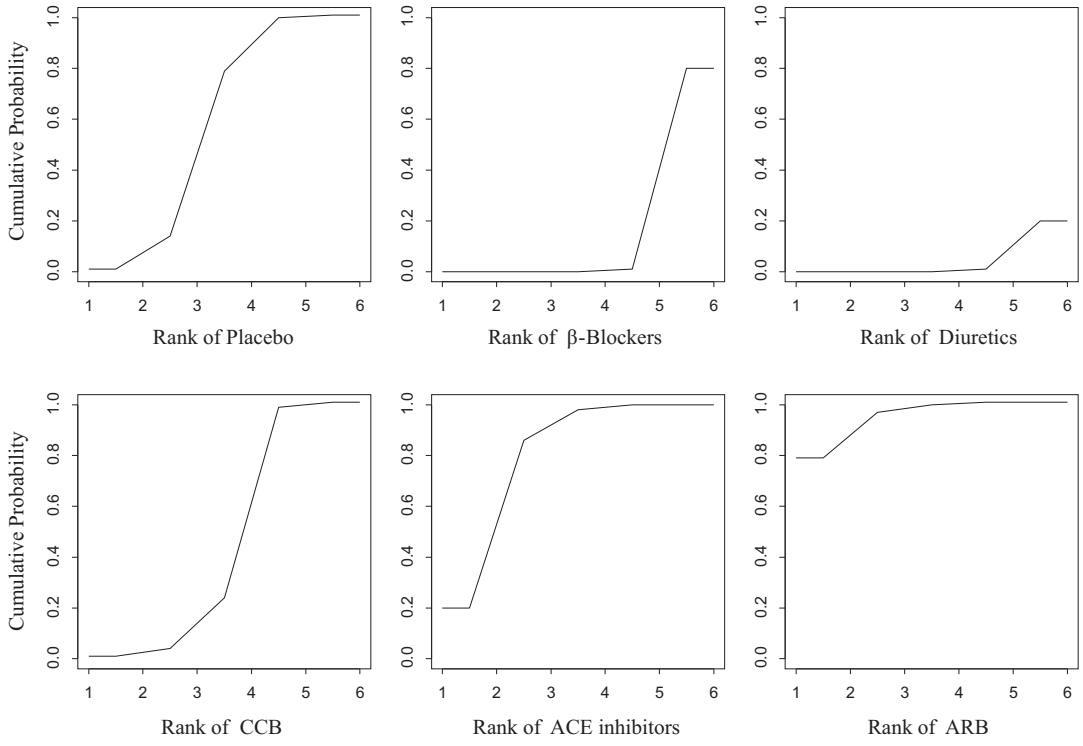


Fig. 10 Plot of area below the cumulative raking curve for all treatments of incident diabetes. Results are based on the consistency hierarchical model (accounting for multi-arm trials)

model is fitted, and the ordering of the treatments according to their effectiveness is collected in each MCMC cycle using the equation:

$$\text{order}_k = \sum_{j=1}^{6} I\left(\mu_{Aj} \leq \mu_{Ak}\right)$$

where $I(\mu_{Aj} \leq \mu_{Ak}) = 1$ if $\mu_{Aj} \leq \mu_{Ak}$ and 0 otherwise, $A = P$ and $j,k = \{BB, D, CCB, ACE, ARB\}$

Then the probability for each treatment $k = \{P, BB, D, CCB, ACE, ARB\}$ of being the jth $(j = 1, \ldots, 6)$ order (P_k^j) is the ratio of MCMC simulations for which $\text{order}_k = j$ over the total number of simulations. Table 15 includes the values of the ranking probabilities and Fig. 9 the corresponding rankograms.

The numerical summary of area below the cumulative raking curve for each treatment k is calculated as $\left(\sum_{j=1}^{5} \text{cum. } P_k^j\right)/5$. The results are presented in Table 16 and the plots in Fig. 10. These results suggest that the best treatment appears to be ARB followed by ACE inhibitors, placebo, CCB, β-blockers, and last diuretics.

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